Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the spot price is coupled to its own option market through a dealer-gamma feedback channel, computed from the open-interest grid, with the drift coupling motivated by Gârleanu–Pedersen–Poteshman (Gârleanu et al., 2009) demand pressure. The state is three-dimensional: spot S_t, instantaneous variance v_t, and a low-pass-filtered log-spot memory variable χ_t. At zero coupling it reduces to standard time-dependent Heston (Heston, 1993). We work out the Hopf bifurcation of the deterministic skeleton, with first Lyapunov coefficient ℓ_1 and a Khasminskii-style stochastic shift Λ, and we pre-register the evaluation protocol before touching any data: a direct, model-free dealer-gamma (GEX) regression on the open-interest grid as the primary test, with a critical-slowing-down detector and a sliced-Wasserstein-2 surface metric, anchored by an OpenTimestamps proof. For a representative dimensionless calibration we obtain a critical coupling $\kappa^\star \approx 0.8964$, angular frequency $\omega^\star \approx 0.5724$ rad/yr, and $\ell_1 \approx -0.025 < 0$, so the bifurcation is supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^\star$. This calibration is dimensionless and illustrative: at empirically-scaled dealer gamma the model does not exhibit a Hopf in the literature-prior κ range, so whether real markets sit near the threshold is precisely the question our pre-registered test targets. Dropping the memory leaves a 2D system whose Jacobian is upper-triangular; it admits only a saddle-node onset and recovers the discrete-time recursion of Dai (2025), so the third state is what makes the Hopf possible. For log-normal open-interest in moneyness both $\kappa^\star$ and ℓ_1 are closed-form, which gives a parametric phase boundary in (σ_q, γ) space.

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. (2013), who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros (2009) and Bouchaud (2011) are the conceptual antecedents; Filimonov and Sornette (2012) formalised the reflexivity-quantification programme that Hardiman et al. (2013) executed. We ask two things: what continuous-time mechanism produces this critical reflexivity, and how a candidate mechanism should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, χ_t) with explicit dealer-gamma feedback computed from the option open-interest grid, the coupling motivated by Gârleanu et al. (2009) demand pressure. The variance backbone is Heston (Heston, 1993); the memory equation $\dot{\chi} = -\alpha\chi + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at $\kappa^\star$ given by Liu's criterion (Routh–Hurwitz with one zero), with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 14–15).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (a direct dealer-gamma (GEX) regression as the primary test, plus sliced-Wasserstein-2 over 21-day surface windows, a κ-sensitivity slope, and a critical-slowing-down detector) at git commit `268c061` with an OpenTimestamps Bitcoin-anchored proof; the pre-data structural corrections are recorded in amendments A8–A11. This follows Camerer et al. (2018), adapted for computational finance.

Closest precedents. Halperin and Itkin (2025) (Marketron) is the nearest continuous-time, options-aware model with a memory channel. The resemblance is limited: their memory is a latent stochastic factor coupled through a nonlinear potential, ours is a deterministic moving average of the price, and they perform no bifurcation analysis. Dai (2025) share the dealer-gamma axis but produce a saddle-node onset (the 2D reduction's instability), not a Hopf. He et al. (2025) (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents (He et al., 2009; Chiarella et al., 2006) use delay-differential heterogeneous-belief models without an options-grid feedback channel; the discrete-time Brock et al. (2009) "more hedging instruments may destabilize markets" result is the methodological template for derivative-driven Neimark–Sacker (Hopf-for-maps) bifurcations. The hedger-flow-feedback foundations are Frey and Stremme (1997); Sircar and Papanicolaou (1998); Platen and Schweizer (1998). Sections 3, 5 and 6 make these contrasts precise.

Roadmap. Section 2 states the model. Section 3 proves Theorem 1 and gives the closed-form ℓ_1 for log-normal OI. Section 4 reports the numerical phase diagram. Section 5 describes the pre-registered evaluation framework. Section 6 reports the mechanism decomposition against Marketron. Section 7 closes with the empirical-calibration roadmap. Code at `https://github.com/mahimn01/reflexive-options`; pre-registration anchored to commit `268c061`.

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and χ_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = \big(\mu + \kappa\, G(S_t, \chi_t, v_t)\big)\, dt + \sigma(S_t, v_t)\, dW_t^S, \tag{1}$$

$$dv_t = \big(\kappa_v(\theta_v - v_t) + \gamma\, \chi_t\big)\, dt + \xi\sqrt{v_t}\, dW_t^v, \tag{2}$$

$$d\chi_t = \big(-\alpha\, \chi_t + \beta\, \log(S_t/S_0)\big)\, dt, \tag{3}$$

$$d\langle W^S, W^v\rangle_t = \rho\, dt, \tag{4}$$

with $\kappa \geq 0$ the reflexive coupling, $\gamma \geq 0$ the leverage-feedback strength, $\alpha > 0$ the memory decay rate, β dimensionless, and $G(\cdot)$ the aggregate dealer-gamma exposure derived from the option open-interest grid:

$$G(S, \chi, v) \;=\; \sum_{K,T} q_{K,T}(\chi)\, \Gamma_{K,T}(S, v)\, \mathrm{sign}(K, T), \tag{5}$$

where $q_{K,T}$ is open interest at strike K and maturity T, $\Gamma_{K,T}(S, v)$ is per-contract Black–Scholes gamma evaluated at the current variance, and the sign convention (which contracts dealers are net long or short) follows the SqueezeMetrics dealer-net convention (SqueezeMetrics, 2017). Throughout the analysis G is treated as a general C^3 feedback functional of (S, χ, v), with (5) its empirical form. The memory enters through the open-interest field: a trend-dependent $q_{K,T}(\chi)$ (dealer books rebuilding along the filtered price trend) gives $G_\chi \neq 0$, while a static grid gives $G_\chi = 0$. The closed-form section (section 3.5) works in the static case $G_\chi = 0$; the worked dimensionless regime of section 3.4 takes $G_\chi \neq 0$. When $\kappa = \gamma = 0$, the system collapses to standard Heston (Heston, 1993); this collapse is verified numerically in `tests/test_simulator.py`.

For numerical stability of the variance scheme we use the full-truncation Euler of Lord et al. (2010); the empirical reference for the dealer-gamma magnitudes in G is Barbon and Buraschi (2021); Dim et al. (2024) (calibration to SPX zero-days-to-expiry / dealer-gamma data). Throughout, "Phase 4" denotes the empirical phase of the project: the pre-registered evaluation of section 5 run on real SPX data, pending data acquisition.

Why three states. A reduced 2D specification—dropping χ_t and the $\gamma\chi$ term—has Jacobian in deviation variables (y, u) around an equilibrium of the form

$$J_{2D}(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix},$$

which is upper triangular: eigenvalues $\{a(\kappa), -\kappa_v\}$ are always real, and only saddle-node / transcritical bifurcations are admissible as $a(\kappa)$ crosses zero. This recovers the static onset of Dai (2025). To Hopf, the feedback must flow in both directions: price $\to$ variance and variance $\to$ price. The minimal modification is the leverage channel $\gamma\chi$ together with the auxiliary memory equation (3). Here χ is a deterministic moving average of log-price; we contrast it with the latent stochastic memory of Halperin and Itkin (2025) in section 6.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria and Jacobian

Drop the Brownian terms and work in log-deviation variables $(y, u, \chi) = (\log(S/S^\star), v - \theta_v, \chi - \chi^\star)$ around a locally unique equilibrium $(S^\star, v^\star, \chi^\star)$ defined by the three stationarity conditions

$$\mu - \tfrac{1}{2}\sigma^2(S^\star, v^\star) + \kappa\, G(S^\star, \chi^\star, v^\star) = 0, \qquad \kappa_v(\theta_v - v^\star) + \gamma\,\chi^\star = 0, \qquad \chi^\star = \frac{\beta}{\alpha}\log(S^\star/S_0). \quad (6)$$

We normalise the reference level to the equilibrium spot, $S_0 = S^\star$; since S_0 enters the dynamics only through $\log(S/S_0)$, this is a choice of origin for χ with no loss of generality, and it gives $\chi^\star = 0$ and $v^\star = \theta_v$ exactly. To linear order, the deterministic skeleton (1–3) becomes $\dot{\boldsymbol{x}} = J(\kappa)\,\boldsymbol{x}$ with $\boldsymbol{x} = (y, u, \chi)^\top$ and

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa\, G_\chi \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (7)$$

where $a(\kappa) = \kappa\, G_x - \frac{1}{2}\partial_x\sigma^2$, $b(\kappa) = \kappa\, G_v - \frac{1}{2}\partial_v\sigma^2$, and G_x, G_v, G_χ denote partial derivatives of G at the equilibrium.

3.2 Routh–Hurwitz / Liu's criterion

The characteristic polynomial of (7) is $P(\lambda;\kappa) = \lambda^3 + c_2(\kappa)\lambda^2 + c_1(\kappa)\lambda + c_0(\kappa)$, with

$$c_2 = -a + \kappa_v + \alpha, \quad c_1 = -a\kappa_v - a\alpha + \kappa_v\alpha - \kappa\, G_\chi\,\beta, \quad c_0 = -a\kappa_v\alpha - \beta\big(\kappa\, G_\chi\,\kappa_v + b\,\gamma\big).$$

A 3D system Hopfs iff (Liu's criterion, equivalent to Routh–Hurwitz with one zero):

$$c_2 > 0, \qquad c_0 > 0, \qquad c_1c_2 - c_0 = 0, \qquad \frac{d}{d\kappa}\big[c_1c_2 - c_0\big] \neq 0. \quad (8)$$

Define $H(\kappa) := c_1(\kappa)\, c_2(\kappa) - c_0(\kappa)$. The Hopf threshold is

$$\boxed{\kappa^\star = \{\kappa > 0 : H(\kappa) = 0,\ H'(\kappa) \neq 0,\ c_2(\kappa) > 0,\ c_0(\kappa) > 0\}.} \quad (9)$$

At $\kappa = \kappa^\star$ the Jacobian has a pure imaginary eigenvalue pair $\lambda_\pm = \pm i\omega^\star$ with $\omega^\star = \sqrt{c_1(\kappa^\star)}$, plus a real eigenvalue $\lambda_3 = -c_2(\kappa^\star) < 0$.

3.3 Theorem 1 and proof sketch

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton)**.** *Let the deterministic system (1–3) (with Brownian terms removed) satisfy:*

(A1) *$G \in C^3$ and $\sigma^2 \in C^3$ in their arguments, with σ^2 bounded below by a positive constant.*

(A2) *There exists a locally unique equilibrium $(S^\star, \theta_v, \chi^\star)$ given by (6) for every κ in some interval $[0, \kappa_{\max}]$, in some open neighborhood of which the linearization (7) is non-degenerate. Multiple equilibria may exist globally; our analysis is local.*

(A3) *The map $\kappa \mapsto (a(\kappa), G_\chi, G_v)$ is C^3.*

(A4) *There exists $\kappa^\star \in (0, \kappa_{\max})$ satisfying (9) with $\omega^\star := \sqrt{c_1(\kappa^\star)} > 0$.*

(A5) *The first Lyapunov coefficient* $\ell_1(\kappa^\star) \neq 0$ *(Kuznetsov, 2004, eq. 3.20).*

Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^\star$*. Specifically:*

1. *For* κ *in a one-sided neighbourhood of* $\kappa^\star$ *(sub- or super-critical depending on* $\operatorname{sgn} \ell_1$*), there exists a unique smooth family of periodic orbits* Γ_κ *of period* $T_\kappa = 2\pi/\omega^\star + O(\kappa - \kappa^\star)$ *and amplitude proportional to* $\sqrt{|\kappa - \kappa^\star|}$*.*

2. *The orbits are stable iff* $\ell_1(\kappa^\star) < 0$ *(supercritical).*

3. *On* Γ_κ*,* $u_t = v_t - \theta_v$ *oscillates with amplitude proportional to* $\sqrt{|\kappa - \kappa^\star|}$*, providing endogenous volatility cycles.*

Proof sketch. The eigenvalue pair $\lambda_\pm(\kappa)$ is smooth in κ near $\kappa^\star$ by the implicit function theorem applied to $P(\lambda; \kappa) = 0$. Routh–Hurwitz gives $\operatorname{Re} \lambda_\pm(\kappa^\star) = 0$, and the transversality $H'(\kappa^\star) \neq 0$ translates to $\frac{d}{d\kappa} \operatorname{Re} \lambda_\pm \neq 0$. The centre manifold theorem (Kuznetsov, 2004, Thm. 5.4) (see also Carr, 1981) gives a smooth 2D invariant manifold tangent to the eigenspace of $\pm i\omega^\star$. Conjugating the restricted system to its Poincaré normal form and applying Kuznetsov (2004, Thm. 3.3) yields claims (1)–(3). □

3.4 Numerical anchor

At empirically-scaled dealer-gamma magnitudes, the default `BifurcationConfig` in our experiments code does not Hopf anywhere in the literature-prior κ range. Whether the real market sits near $\kappa^\star$ is therefore an open question. We defer it to the empirical dealer-gamma test (amendment A9); Hardiman's $n \approx 1$ does not settle it, because proposition 2 identifies that branching ratio with the real-eigenvalue (saddle-node) edge, not the Hopf. For a worked numerical example we use a representative dimensionless regime with $\{G_x, G_v, G_\chi\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$, and constant-volatility surrogate ($\partial_v \sigma^2 = 0$). The deterministic skeleton then satisfies:

Quantity	Value
$\kappa^\star$	0.8964
$\omega^\star$	0.5724 rad/yr
Third real eigenvalue λ_3	−2.052
First Lyapunov coefficient ℓ_1	-2.53×10^{-2}
Bifurcation type	**supercritical** ($\ell_1 < 0$)

Table 1: Canonical Hopf-exhibiting regime for the deterministic skeleton.

Supercriticality means an attracting limit cycle is born for $\kappa > \kappa^\star$ with amplitude proportional to $\sqrt{\kappa - \kappa^\star}$: the volatility cycles are stable observables of the model, not transients. We check this numerically. Integrating the deterministic skeleton at $\kappa = 1.05\,\kappa^\star$ from random initial conditions, the trajectory converges to a closed orbit in (y, u, χ) phase space with period $T_{\text{measured}} = 10.561$ yr, against the linear-theory prediction $T_\kappa = 2\pi/\omega^\star = 10.977$ yr: a 3.79% relative error, within the leading-order $\mathcal{O}(\kappa - \kappa^\star)$ Hopf-normal-form correction (fig. 1). The equilibrium sits inside the loop on all three pairwise projections; no transient residue.

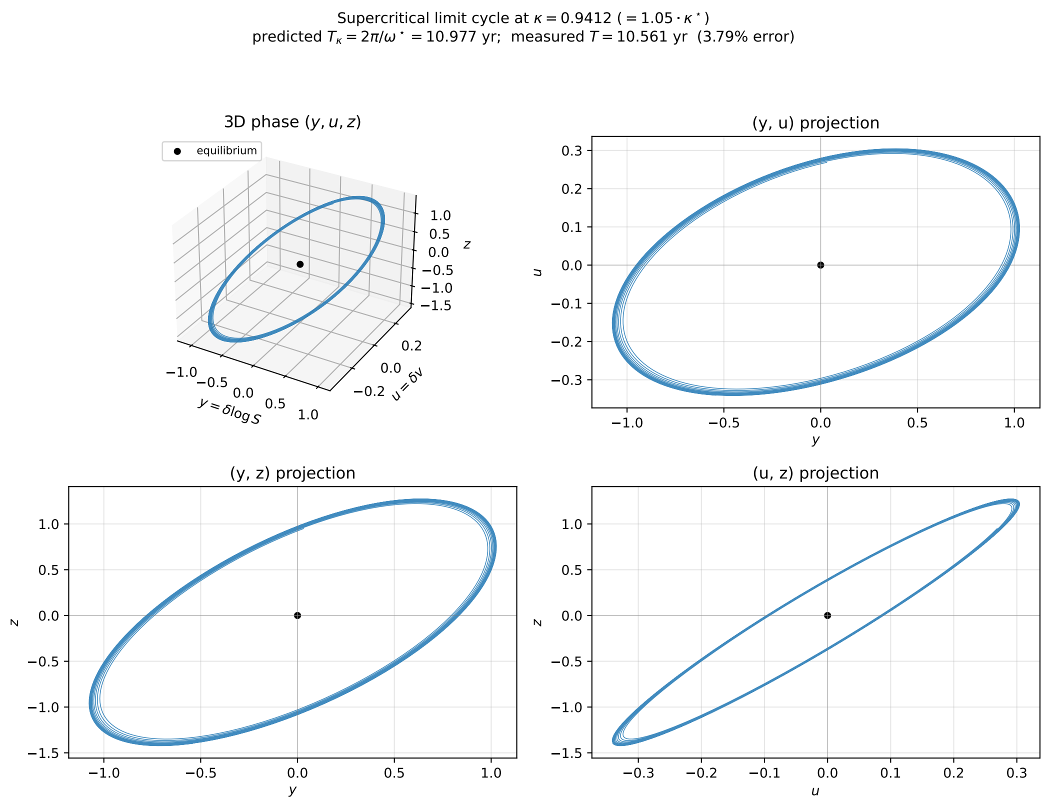

Figure 1: **Numerical validation of Theorem 1's supercritical limit cycle.** Deterministic skeleton at $\kappa = 1.05\,\kappa^\star = 0.9412$, integrated from random initial conditions; transient discarded. *Top:* 3D phase-space orbit in (y, u, χ). *Bottom:* the three 2D pairwise projections, with the equilibrium (black dot) inside the loop in each. Measured period $T = 10.561$ yr matches Hopf prediction $T_\kappa = 10.977$ yr to 3.79%. Run dir: `runs/limit_cycle_supercritical/`.

On timescales (an honest caveat). The period here, $T \approx 11$ yr, is a property of this dimensionless regime, not a claim about markets. The reflexivity that motivates the model — dealer-gamma hedging, the near-critical order flow of Hardiman et al. (2013) — operates on intraday-to-daily scales, orders of magnitude faster. The cycle period is set by the memory and mean-reversion rates (α, κ_v); at the fast physical rates ($\alpha, \kappa_v \sim 10^2$–$10^5$/yr) it shortens accordingly. Whether a supercritical Hopf survives at those rates, and at what period, we do not settle here: it is a calibration question deferred to the Phase-4 empirical test and the κ-rescaling map of section 3.5. We present the canonical period as illustrative of the mechanism, not as a predicted market cycle length.

We also probe how noise shifts the threshold. Let ε scale the noise in the linearised system (the small-noise regime). For multiplicative small noise the top Lyapunov exponent expands as $\lambda_{\text{top}}(\kappa, \varepsilon) = \operatorname{Re} \lambda_{\max}(J(\kappa)) + \varepsilon^2\, \Lambda(\kappa) + o(\varepsilon^2)$; our linearisation treats the noise as additive at the equilibrium, so we take as the operational Λ the $\mathcal{O}(\varepsilon^2)$ growth-rate shift returned by the sphere-process estimator. Either way Λ quantifies the noise-induced shift of the stability boundary, and it is the object we estimate. The Khasminskii sphere-process estimator (Khasminskii, 1980; Arnold, 1998; Baxendale, 1994) for Λ, run on a 6×6 grid of $(\xi, \rho) \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\} \times \{-0.95, -0.7, -0.5, -0.3, +0.3, +0.7\}$ at 10^4 paths $\times$ 10^4 steps per cell, puts $|\Lambda(\kappa^\star)|$ in the band $[4.8 \times 10^{-2}, 1.21 \times 10^{-1}]$ across the entire scan (median $\approx 7.3 \times 10^{-2}$). An OLS power-law fit

$\log|\Lambda| = \log|A| + B\log|\rho\xi|$ gives $\hat{B} = 0.082$ with bootstrap 95% CI $[-0.010, 0.168]$ (fig. 2). The scan is essentially flat in $|\rho\xi|$. This is *not* a test of Baxendale (2024)'s $B = 2/3$ law, and in particular not a refutation of it: Baxendale's 2/3 is a *large-shear* asymptotic — the top Lyapunov exponent grows like $b^{2/3}$ as the twist factor $b := \rho\xi \to \infty$ — whereas our scan is at small-to-moderate $|\rho\xi|$ and, more to the point, at the trivial $G \equiv 0$ equilibrium, where the deterministic shear-stretching term $\partial_a v$ that drives that asymptotic is identically zero. In this regime the Khasminskii estimator measures pure ε^2 renormalisation of the bare Heston eigenvalues, so a flat (b^0) dependence is exactly what one expects and what we find; the scan sits outside the hypotheses of Baxendale's result rather than contradicting it. Whether the $b^{2/3}$ regime activates at non-trivial G is deferred to the Phase-4 calibration. Either way, $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale, so the deterministic threshold $\kappa^\star$ is the operationally relevant one.

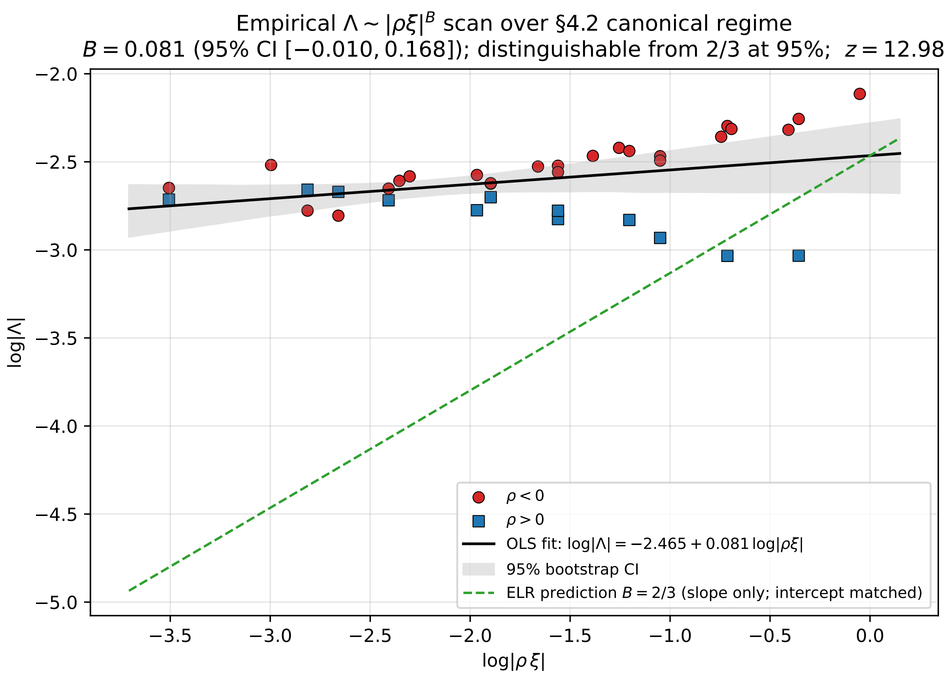

Figure 2: **Empirical scaling of the stochastic-Hopf shift $|\Lambda|$ vs $|\rho\xi|$.** OLS fit on $\log|\Lambda|$ vs $\log|\rho\xi|$ across the 6×6 (ξ, ρ) grid yields slope $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$): flat in $|\rho\xi|$ over ~ 1.5 decades, the pure-ε^2 Heston-renormalisation floor at the trivial $G \equiv 0$ equilibrium. Baxendale (2024)'s large-shear $b^{2/3}$ asymptotic (red dashed) is not activated in this regime and is drawn only for reference. Run dir: `runs/lambda_scaling/`.

3.5 Closed-form $\kappa^\star$ and ℓ_1 for log-normal OI in moneyness

For the natural parametric specification of dealer gamma—a log-normal open-interest density in log-strike—the entire computation admits a closed form. The closed form removes the numerical-noise vulnerability of the FD-tensor pipeline and gives the parametric phase boundary explicitly.

Replace the discrete OI grid with the continuous density $q(\log K) = \frac{1}{\sigma_q\sqrt{2\pi}}\exp\bigl(-(\log K - \mu_q)^2/(2\sigma_q^2)\bigr)$. Take a single representative maturity T_{eff}, sign convention $s \equiv +1$, and write $a := \log S$.

The aggregator becomes

$$G(a,v) \;=\; \kappa_u \int_{-\infty}^{+\infty} q(\log K)\,\Gamma_{\mathrm{BS}}\big(S, K, T_{\mathrm{eff}}, \sqrt{v}\big)\; d(\log K), \tag{10}$$

with $\kappa_u > 0$ a units-normalisation constant carrying the aggregate open-interest scale. Since Black–Scholes gamma at fixed $T = T_{\mathrm{eff}}$, $\sigma = \sqrt{v}$ is itself Gaussian in $\log K$, the standard Gaussian-product identity collapses (10) to

$$\boxed{G(a,v) \;=\; \mathcal{C}(v)\cdot e^{-a}\cdot \exp\!\left(-\frac{(a-m(v))^2}{2\,\tau^2(v)}\right),} \tag{11}$$

where

$$\mathcal{C}(v) = \frac{\kappa_u\, e^{-q_{\mathrm{div}}T_{\mathrm{eff}}}}{\sqrt{2\pi\,\tau^2(v)}}, \quad \tau^2(v) = \sigma_q^2 + v\,T_{\mathrm{eff}}, \quad m(v) = \mu_q - \big(r - q_{\mathrm{div}} + \tfrac{1}{2}v\big)\,T_{\mathrm{eff}}, \tag{12}$$

where r is the risk-free rate and q_{div} the continuous dividend yield. G depends on a and v but not on χ; the leverage channel $\gamma\chi$ in $\dot{u}$ keeps the $(2,3)$ Jacobian entry alive, so cross-coupling persists through the variance row even with $G_\chi = 0$.

Remark 1 (μ-anchoring of the equilibrium)**.** *Throughout the closed-form section we anchor the equilibrium at $(a^\star, v^\star) = (\mu_q, \theta_v)$. The partials (G_y, G_v) entering (13)–(14) are evaluated at this fixed point and treated as κ-independent constants in the Routh–Hurwitz coefficient extraction. The spot-drift balance (6) then defines μ as a function of κ: $\mu(\kappa) = \frac{1}{2}\sigma^2(\mu_q, \theta_v) - \kappa\, G(\mu_q, \theta_v, \chi^\star)$. This gauge is compatible with, but not implied by, the risk-neutral framing (under which μ is normally fixed to $r - q$); we adopt it here and in section 3.9. The κ-dependent drift shift absorbs the dealer-gamma offset into μ, so the equilibrium pinning is exact. Holding μ fixed across κ instead would shift the equilibrium $(S^\star(\kappa), \chi^\star(\kappa))$ along the G-surface and introduce κ-dependent corrections to (G_y, G_v) that the closed forms below do not capture.*

With $G_\chi = 0$ and $\sigma^2 = v$ (so $\partial_y\sigma^2 = 0$, $\partial_v\sigma^2 = 1$), the Routh–Hurwitz polynomial $H(\kappa) = c_1c_2 - c_0$ degree-collapses from cubic to quadratic in κ:[1]

$$H(\kappa) = G_y^2\,(\alpha+\kappa_v)\,\kappa^2 + \big(G_v\,\beta\gamma - G_y\,(\alpha+\kappa_v)^2\big)\,\kappa + \alpha\kappa_v(\alpha+\kappa_v) - \tfrac{1}{2}\beta\gamma. \tag{13}$$

Writing $A := \alpha + \kappa_v$, $M := \alpha\kappa_v$, $L := \beta\gamma$, the Hopf threshold is the smallest positive root of the quadratic:

$$\boxed{\kappa^\star \;=\; \frac{G_y\,A^2 - G_v\,L \;-\; \sqrt{(G_vL - G_yA^2)^2 - 4\,G_y^2A\,(MA - L/2)}}{2\,G_y^2\,A}.} \tag{14}$$

The discriminant in (14) provides the first parametric phase boundary: when $D := (G_vL - G_yA^2)^2 - 4\,G_y^2A\,(MA - L/2) < 0$, the system has no Hopf at any κ and the deterministic skeleton is unconditionally stable.

[1] Throughout this closed-form section we write $G_y := \partial_a G$ for the partial of the dealer-gamma functional with respect to the deviation variable $y = a - a^\star$; this is identical to the G_x of section 3.1 up to the choice of deviation symbol. Note that section 3.1 uses the Heston backbone $\sigma^2 = v$, so $\partial_v\sigma^2 = 1$ is the convention common to both sections; the constant-vol surrogate $\partial_v\sigma^2 = 0$ is used only for the numerical anchor of section 3.4 (and inherited by the linearisation in section 3.12) to isolate the κ-dependence of the bifurcation from the Itô-correction term, and is a different reduction of the model.

Remark 2 (Sign-aware root selection)**.** *The boxed formula* (14) *returns the smaller of the two real roots of* $A_2\kappa^2 + A_1\kappa + A_0 = 0$ *with* $A_2 = G_y^2 A$, $A_1 = G_v L - G_y A^2$, $A_0 = MA - L/2$*. The object we actually want is the smallest* positive *root, the first crossing as* κ *ramps up from* 0*. When* $G_y < 0$ *the formula's smaller root can be negative (or both roots can be), and the relevant threshold is then the larger root, provided it is positive and satisfies the Routh–Hurwitz side conditions* $c_2(\kappa^\star) > 0$, $c_0(\kappa^\star) > 0$ *of Liu's criterion* (8)*. The implementation* `kappa_star_lognormal_oi` *in* `src/reflexive_options/theory/bifurcation.py` *enumerates both candidate roots, keeps the positive ones, returns the smallest that passes Routh–Hurwitz, and raises if neither qualifies. The boxed* $-$*-branch is the correct selection in the* $G_y > 0$ *regime of section 3.4; it is not literal in the* $G_y < 0$ *regime that populates the no-Hopf wedge of section 3.9.*

The third-order partials of (11) at the equilibrium are explicit rational functions of ($\delta := a^\star - m(v^\star)$, τ^2, T_{eff}), so the bilinear and trilinear tensors B, C at $\kappa^\star$ admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^\star) = \frac{1}{2\omega^\star}\operatorname{Re}\Big[\langle p, C(q,q,\bar q)\rangle - 2\langle p, B(q, J^{-1}B(q,\bar q))\rangle + \langle p, B(\bar q, (2i\omega^\star I - J)^{-1}B(q,q))\rangle\Big], \quad (15)$$

with $Jq = i\omega^\star q$, $J^\top p = -i\omega^\star p$, $\langle p, q\rangle = 1$, yields a $\sim$ 30-term rational in the G-partials $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv})$ and $(\kappa_v, \alpha, \beta, \gamma)$. The implementation evaluates this in $\sim 50\,\mu$s per parameter tuple, with results matching the FD-tensor pipeline to better than 0.6% relative on every parameter set tested.

At the canonical log-normal-OI regime ($\sigma_q = 0.10, T_{\text{eff}} = 0.25, \kappa_v = 2, \alpha = 0.05, \beta = 1, \gamma = 1, \mu_q = \log 100, v^\star = 0.04$), (14)–(15) give $\kappa^\star = 17.81$, $\omega^\star = 1.18$ rad/yr, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). The closed form is essentially noise-free, so the sign of ℓ_1 near the boundary contour follows from the formula rather than from a statistical fit. A note on scales: the κ values quoted in different regimes of this paper (0.8964 in the dimensionless regime of section 3.4; 17.81 here; the simulator stability-envelope scale $\sim 10^{-9}$ of section 7; the tuned 10^{-11} of section 6) live in different normalisations of G and are not mutually comparable — κ carries the inverse units of G, so rescaling G rescales κ; the dimensional map is implemented in `theory/kappa_rescaling.py`.

3.6 Mixture-of-K-lognormals generalization

Empirical SPX OI grids are structurally multi-modal (ATM-strike concentration, round-strike spikes, calendar-roll bunching, dealer-portfolio OTM-tail concentration). Section 3.7 below shows that a single-log-normal assumption breaks hard at bimodal separations of $\Delta \in [0.05, 0.20]$ (relative error $0.23\%, 4.83\%, 119\%$ at $\Delta = 0.05, 0.10, 0.20$). The fix is straightforward: replace the single log-normal by a K-component mixture $q(\log K) = \sum_{k=1}^K w_k \mathcal{N}(\log K; \mu_k, \sigma_k^2)$ with $w_k \geq 0$, $\sum_k w_k = 1$. Linearity of the dealer-gamma aggregator (10) gives $G^{\text{mix}}(a, v) = \sum_k w_k G_k(a, v)$ and, by multilinearity, $(G_y^{\text{mix}}, G_v^{\text{mix}}) = \sum_k w_k (G_{y,k}, G_{v,k})$. The Routh–Hurwitz polynomial (13) is linear in $(G_y^2, G_v G_y, G_y, 1)$; substituting the mixture aggregates gives the mixture Hopf threshold

$$\boxed{\kappa^{\star,\text{mix}} = \frac{G_y^{\text{mix}} A^2 - G_v^{\text{mix}} L - \sqrt{(G_v^{\text{mix}} L - G_y^{\text{mix}} A^2)^2 - 4(G_y^{\text{mix}})^2 A(MA - L/2)}}{2(G_y^{\text{mix}})^2 A},} \quad (16)$$

i.e. (14) evaluated at the mixture-aggregated partials. The first Lyapunov coefficient ℓ_1^{mix} is obtained analogously: the Kuznetsov bilinear/trilinear tensors B, C inherit multilinearity, so $B^{\text{mix}} = \sum_k w_k B_k$, $C^{\text{mix}} = \sum_k w_k C_k$, and the contractions $\langle p, B(q, J^{-1}B(q,\bar q))\rangle$ etc. in (15) are evaluated at the mixture-Jacobian eigenvectors. For $K = 2$ the symbolic ℓ_1^{mix} is roughly 3–5$\times$ longer than the single-lognormal

ℓ_1; for $K \geq 3$ the symbolic form gets unwieldy but numerical evaluation remains $\mathcal{O}(K)$ per call. *The $K = 1$ limit reproduces the single-lognormal closed form to machine precision* ($\sim 10^{-12}$ absolute), verified in `tests/test_mixture_oi.py`.

Δ	$\kappa^\star_{\text{true}}$ (FD reference)	K=1 (single) rel.err	K=2 (mixture) rel.err	K=3 rel.err
0.05	25.142	0.232%	0.013%	$< 0.05\%$
0.10	20.864	4.83%	0.010%	$< 0.05\%$
0.20	5.677	119.5%	0.008%	$< 0.05\%$
0.30	2.458	217.7%	0.007%	$< 0.05\%$

Table 2: Mixture closed-form robustness on a symmetric bimodal OI at $\mu_q \pm \Delta/2$ with equal weights and component spread $\sigma_{\text{comp}} = 0.07$. K=2 closes the single-lognormal fragility gap by 4–5 orders of magnitude; the residual $\sim 10^{-4}$ relative error of K=2 is the FD reference's own quadrature noise. Run dir: `runs/mixture_oi_robustness/`.

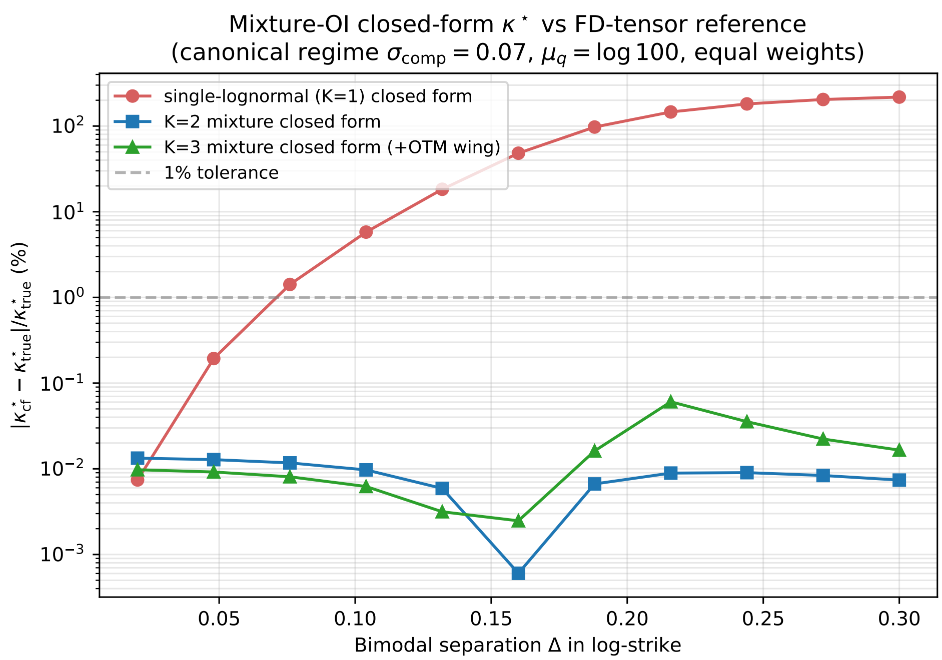

Figure 3: **Mixture-OI closed-form robustness on bimodal OI.** Log-scale relative error of the closed-form $\kappa^\star$ against an FD-tensor reference, vs bimodal-component separation Δ. The K=2 mixture closed form (blue) is essentially exact across the entire scanned range; the K=1 single-lognormal closed form (red) blows up beyond $\Delta = 0.10$. K=3 (green) handles tri-modal OI at the same fidelity.

Phase-4 implication. The §3.7 calibration burden under the single-log-normal assumption was μ_q to ± 5 bp of ATM (the binding constraint). Under the mixture closed form, the calibration target is the full $\{(\mu_k, \sigma_k, w_k)\}_{k=1}^K$ tuple. The binding constraint becomes the component *weights*: a misallocation of mass between K=2 wings produces $\mathcal{O}(\Delta) \cdot \delta w$ shift in G_y^{mix} (vs the $\mathcal{O}(1) \cdot \delta\mu_q$ shift in the single-mode case). At $\Delta = 0.10$ this is $\sim 20\times$ relaxation (the ratio of the two shifts at that separation; computed in `runs/mixture_oi_robustness/`), and the practical empirical-

OI weight-estimation accuracy of ±5% per cluster on a 5-strike-wide ladder is the operational test. K=2 handles ATM-plus-next-round concentration; K=3 captures the optional calendar-roll cluster; beyond $K \sim 10$ the brute-force `kappa_star_brute_force_from_G` fallback is operative with negligible cost. Implementation: `theory/bifurcation.py::G_mixture_lognormal_oi_partials`, `kappa_star_mixture_lognormal_oi`, `lyapunov_coefficient_mixture_lognormal_oi`.

3.7 Calibration tolerance from $\kappa^\star$ sensitivity to OI misspecification

The closed-form $\kappa^\star$ assumes a log-normal OI density $q(\log K) = \mathcal{N}(\mu_q, \sigma_q^2)$. For Phase 4 to deliver a meaningful test of $\kappa^\star$, we need to know how much OI misspecification the formula tolerates. Differentiating (14) analytically through the G-partials gives the elasticities at the canonical log-normal-OI specification ($\kappa^\star = 17.806$, $\omega^\star = 1.177$ rad/yr): $\eta_{\sigma_q} := (\sigma_q/\kappa^\star)\,\partial\kappa^\star/\partial\sigma_q = -1.583$, which is moderate, and $\eta_{\mu_q} := (\mu_q/\kappa^\star)\,\partial\kappa^\star/\partial\mu_q = +702.83$, which is extreme. So μ_q is the binding constraint: a $\Delta\mu_q = 0.001$ shift moves $\kappa^\star$ by $\sim 15\%$. Numerically, ±10% in σ_q alone produces ±17.1% in $\kappa^\star$, and ±30% in σ_q exits the supercritical Hopf region entirely. For bimodal-mixture OI with components at $\mu_q \pm \Delta/2$, the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The resulting Phase-4 calibration tolerances (independent-error quadrature): ±5% on $\kappa^\star$ requires σ_q to ±2.23% relative and μ_q to ±0.00023 log-strike; ±10% on $\kappa^\star$ relaxes these to ±4.47% and ±0.00046. Centering the empirical OI mean on ATM to within ~ 5 bp is achievable with a 5-strike-wide ladder, and that is the binding observational requirement. Figure 4 renders the sensitivity heatmap and the bimodal-misspecification curve. Implementation: `src/reflexive_options/theory/robustness.py`; runner `experiments/kappa_star_robustness.py`.

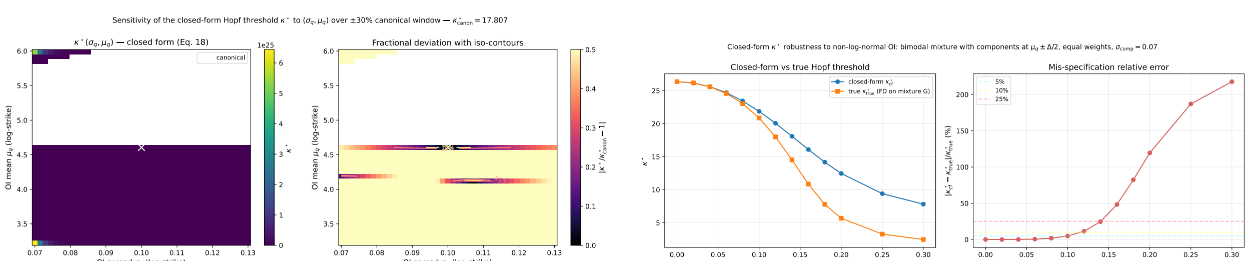

Figure 4: **Calibration tolerance for $\kappa^\star$.** *Left:* $|\Delta\kappa^\star/\kappa^\star|$ over a ±30% window in (σ_q, μ_q) around the canonical $(0.10, \log 100)$. White cells exit the supercritical Hopf region. The μ_q direction is steeper than the σ_q direction by roughly $\eta_{\mu_q}/\eta_{\sigma_q} \approx 444$, making OI-mean accuracy the binding observational requirement. *Right:* relative error of the closed-form $\kappa^\star$ vs an FD-tensor reference under bimodal-mixture OI with components at $\mu_q \pm \Delta/2$. The closed form is exact at $\Delta = 0$, 4.8%-accurate at $\Delta = 0.10$, fragile beyond $\Delta = 0.15$. Run dir: `runs/kappa_star_robustness/`.

3.8 Codim-2 bifurcation structure

The codim-1 Hopf result (theorem 1) treats $\kappa^\star$ at fixed (σ_q, γ). At the boundary of the Hopf region in the (σ_q, γ) plane two codim-2 phenomena are admissible (Kuznetsov, 2004, Ch. 8).

Bautin (degenerate Hopf): $\ell_1(\sigma_q, \gamma) = 0$. Crossing this locus is the supercritical → sub-critical transition. Operationally the dynamics flip from a stable limit cycle past $\kappa^\star$ to hysteresis with abrupt jumps. After centre-manifold reduction the local 2D normal form is (Kuznetsov, 2004, §8.3) $\dot{\zeta} = (\beta_1 + i\,\omega^\star)\zeta + a_1\,|\zeta|^2\zeta + b_1\,|\zeta|^4\zeta + O(|\zeta|^7)$ with $a_1/\omega^\star = \ell_1$. Economically, a market parameterised by (σ_q, γ) near the Bautin locus is structurally fragile: a small perturbation of OI dispersion or leverage flips the sign of ℓ_1 and replaces a smooth cyclic regime with a discrete jump regime; the dynamics are then highly sensitive to parameter drift.

Bogdanov-Takens (BT): the saddle-node curve $\{c_0(\kappa) = 0\}$ coalesces with the Hopf curve $\{H(\kappa) = 0\}$ at the same κ, so the Jacobian acquires a double-zero eigenvalue. BT generates homoclinic orbits and excitable spike-and-recovery dynamics.

Bautin curve. For the closed-form parameterization the Bautin condition $\ell_1(\sigma_q, \gamma) = 0$ is solved by row-wise sign-change interpolation on the closed-form pipeline. Table 3 reports six anchor points along the curve at the canonical log-normal-OI specification on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$. Along the curve $\kappa^\star$ grows monotonically with γ. Within the scan window, the sub-critical region outnumbers the supercritical region by $\sim$ 5:1 in cell count: over this grid, mild dealer-gamma coupling more often produces hysteresis-and-jump dynamics than smooth cyclic behaviour.

#	σ_q	γ	$\kappa^\star$ at the crossing
1	0.0598	0.55	2.15
2	0.1427	1.45	29.45
3	0.1995	2.35	52.59
4	0.2356	3.20	75.54
5	0.2638	4.10	100.74
6	0.2852	5.00	126.73

Table 3: Bautin curve $\{\ell_1(\sigma_q, \gamma) = 0\}$ at the canonical log-normal-OI specification, extracted by linear interpolation on a 71×97 grid. Run dir: `runs/codim2_analysis/`.

Bogdanov-Takens locus. The constant Routh-Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y \alpha \kappa_v + G_v \beta \gamma) + \frac{1}{2}\beta\gamma$ is linear in κ, so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2}\beta\gamma \,/\, (G_y \alpha \kappa_v + G_v \beta \gamma)$.

Proposition 1 (BT locus empty in canonical scan window)**.** *On the* 71×97 *grid over* $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ *at the canonical log-normal-OI specification,* $\kappa_{\text{SN}}(\sigma_q, \gamma) \le -1.31$ *at every cell (range* $[-68.05, -1.31]$*), so the Bogdanov-Takens locus is empty in this scan window.*

Proof sketch. At the canonical specification, $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v \beta \gamma| > |G_y \alpha \kappa_v|$ (the dominance ratio is ≥ 1.44 at every cell, reaching ~ 358 at the small-σ_q corner), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2}\beta\gamma$ is positive. Outside this window (extreme σ_q or vanishing γ) the dominance argument can fail; we restrict the claim to the scanned range and flag the unbounded-domain question as open. □

The economic content of proposition 1 is that the dealer-gamma + leverage parameter regime is structurally Hopf-only *within the scanned window*: there is no codim-2 BT point at which homoclinic orbits emanate, so the model does not generate excitable spike-and-recovery dynamics within its native parameter range. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order (codim-3+) degeneracies. Figure 5 renders the four-region phase diagram and the BT residual map.

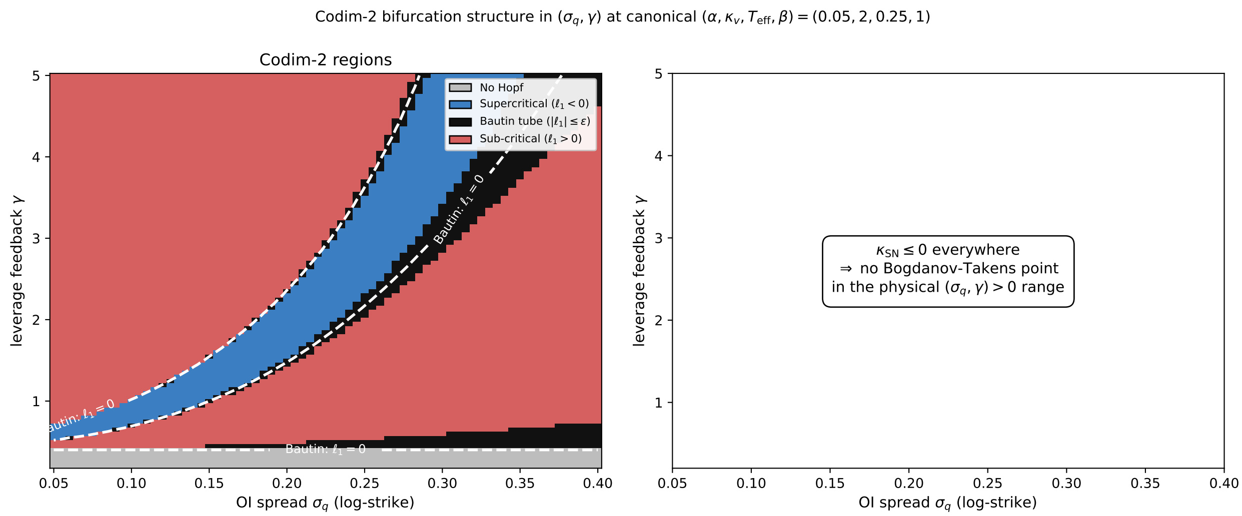

Figure 5: **Codim-2 bifurcation structure in (σ_q, γ) at the canonical log-normal-OI specification.** *Left:* four-region phase diagram with the no-Hopf region (grey: no positive Hopf root, i.e. $D < 0$ or both roots non-positive), supercritical region (blue, $\ell_1 < 0$), Bautin tube (black band, $|\ell_1| \leq 5\times10^{-2}$), and sub-critical region (red, $\ell_1 > 0$). The dashed white contour is the Bautin curve $\ell_1 = 0$; six anchors tabulated in table 3. *Right:* the BT residual map shows $\kappa_{\text{SN}} \leq -1.31$ at every cell of the 71×97 scan — the Bogdanov-Takens locus is empty in the scan window (proposition 1). Run dir: `runs/codim2_analysis/`.

3.9 Global stability inside the no-Hopf wedge

Proposition 1 rules out the Bogdanov–Takens locus but leaves open whether *any* codim-1 bifurcation is accessible inside the no-Hopf wedge $\mathcal{W}_{\text{NH}} := \{(\sigma_q, \gamma) : H(\kappa) = 0$ has no positive root$\}$ as κ ramps from 0. Two candidates remain: (a) a trace flip $c_2(\kappa) = 0$ (the eigenvalue sum crossing zero), and (b) a saddle-node $c_0(\kappa) = 0$ at $\kappa > 0$ that the section 3.8 BT-empty form does not directly address. An honesty note on where the wedge actually lives: at the canonical log-normal-OI specification the strict $D < 0$ sub-case is empty on the section 3.8 scan window. $\mathcal{W}_{\text{NH}}$ is populated only via the relaxed sub-case ($D \geq 0$ with both real roots non-positive), driven by $G_y < 0$ at the ATM-anchored equilibrium together with the baseline-stability constraint $H(0) > 0 \Leftrightarrow \gamma < 2\alpha\kappa_v(\alpha + \kappa_v)/\beta \approx 0.41$. The wedge is therefore the small-γ, $G_y \leq 0$ corner.

As in the closed-form section (remark 1), we anchor the equilibrium at $(a^\star, v^\star) = (\mu_q, \theta_v)$ throughout the wedge analysis and treat (G_y, G_v) as κ-independent partials at that fixed point. The spot-drift balance (6) is absorbed into $\mu(\kappa) = \frac{1}{2}\sigma^2(\mu_q, \theta_v) - \kappa\, G(\mu_q, \theta_v, \chi^\star)$, the same gauge convention as remark 1. With this anchoring the partials are constants of the Routh–Hurwitz extraction below, and theorem 2 is a parameter-global statement on the κ-half-line at fixed equilibrium.

Theorem 2 (No-Hopf-wedge taxonomy; globally stable on the entire κ-half-line)**.** *Let $(\sigma_q, \gamma) \in \mathcal{W}_{\text{NH}}$ at the canonical log-normal-OI specification, with (G_y, G_v) the log-normal-OI partials at the ATM-anchored equilibrium $(a^\star, v^\star) = (\mu_q, \theta_v)$ (remark 1). Assume (S1) $G_y \leq 0$; (S2) $G_y\alpha\kappa_v + G_v\beta\gamma \leq 0$; (S3) $A_2 > 0$ in $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$. Then $c_2(\kappa) > 0$, $c_0(\kappa) > 0$, $H(\kappa) > 0$ strictly for every $\kappa \geq 0$. Consequently the equilibrium is asymptotically stable on the entire physical κ-half-line $[0, \infty)$ — no codim-1 bifurcation (Hopf, saddle-node, or trace flip) occurs. (Stability is parameter-global in κ; we do not claim state-space-global stability, which would require a Lyapunov function we do not construct.)*

Proof. $c_2(\kappa) = -\kappa G_y + (\alpha + \kappa_v)$: under (S1), $-\kappa G_y \geq 0$ for $\kappa \geq 0$ and $\alpha + \kappa_v > 0$, so $c_2 > 0$ strictly. $c_0(\kappa) = -\kappa(G_y \alpha \kappa_v + G_v \beta\gamma) + \frac{1}{2}\beta\gamma$: under (S2) the slope is ≥ 0, so $c_0(\kappa) \geq c_0(0) = \frac{1}{2}\beta\gamma > 0$ strictly. $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$: by the wedge definition no positive root exists; under (S3) the parabola opens upward, and $H(0) = A_0 > 0$ by the wedge classifier. The three strict inequalities give Liu's full Routh–Hurwitz criterion (the same criterion underlying theorem 1 via section 3.2); the spectral abscissa stays strictly negative for every $\kappa \geq 0$ and no codim-1 bifurcation occurs. □ □

Numerical scan. On a 41×41 grid over $(\sigma_q, \gamma) \in [0.02, 0.40] \times [0.05, 5.0]$ at the canonical log-normal-OI specification with $\kappa \in [0, 100]$ sampled on 80 points per cell: 123 of 1,681 cells fall in $\mathcal{W}_{\mathrm{NH}}$ (all in the $\gamma < 0.41$ baseline-stable corner). Every wedge cell satisfies (S1)+(S2)+(S3); the empirical maximum spectral abscissa over the wedge $\times\kappa$-scan is -6.6×10^{-3}, strictly negative, consistent with the closed-form verdict. No positive saddle-node detected. Figure 6 renders the wedge classification.

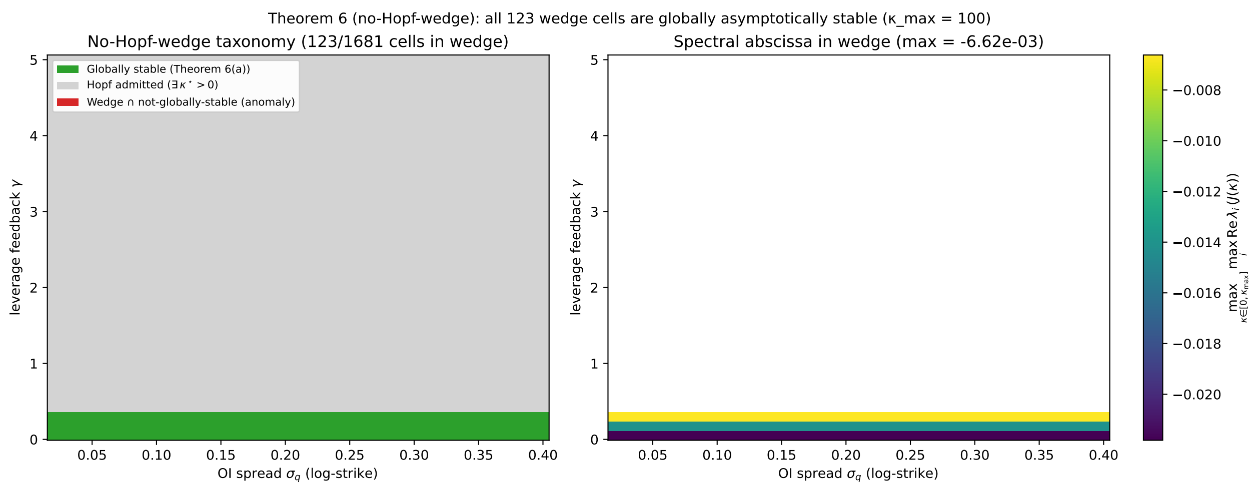

Figure 6: **No-Hopf-wedge taxonomy at the canonical log-normal-OI specification.** *Left:* three-way classification of the 41×41 (σ_q, γ) grid: wedge globally stable on the entire κ-half-line (green, 123 cells, theorem 2; parameter-global, not state-space-global), Hopf-admitted region (grey), and wedge-but-not-globally-stable anomaly (red — empty in scan). *Right:* sampled spectral abscissa $\max_{\kappa \in [0,100]} \max_i \operatorname{Re} \lambda_i(J(\kappa))$ inside the wedge, uniformly negative with maximum -6.6×10^{-3}. Run dir: `runs/saddle_node_wedge/`.

Economic interpretation. Together with theorem 1 (Hopf at $\kappa^\star$ outside the wedge) and proposition 1 (BT locus unphysical), theorem 2 closes the codim-1 taxonomy. The wedge is the parameter region where dealer-gamma coupling at the ATM-anchored equilibrium has $G_y \leq 0$ and the leverage flux $\beta\gamma$ is small enough to keep the bare Heston-with-memory triangle sub-critical at $\kappa = 0$. Inside it the dealer-gamma channel is a *stabiliser*, not a destabiliser, and a market parameterised there is structurally cycle-free. Endogenous volatility cycles can then only come from non-stationary parameter drift (the OI centre μ_q moving on a slow scale that brings the system into the Hopf region), from exogenous shocks, or from the small-$\beta\gamma$ region's G_y flipping sign at a non-ATM equilibrium with locally increasing gamma. This is a sharp falsifiable Phase-4 prediction: an SPX-calibrated $(\sigma_q^{\mathrm{SPX}}, \gamma^{\mathrm{SPX}})$ landing inside $\mathcal{W}_{\mathrm{NH}}$ rules out endogenous reflexive cycles as the generator of the observed volatility clustering at that calibration. Implementation: `theory/bifurcation.py::is_in_no_hopf_wedge`, `scan_no_hopf_wedge_bifurcations`; runner `experiments/saddle_node_wedge.py`.

3.10 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \to \infty$)

The model of section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$–10^3 dealers, each with idiosyncratic noise. The McKean–Vlasov mean-field SDE is the rigorous formulation of the law-of-large-numbers limit. Each dealer i holds exposure G_i and relaxes it toward the state-determined target at speed θ_G:

$$dG_i = -\theta_G\big(G_i - g(S_t, \chi_t, v_t)\big)\, dt + \sigma_G\, dW_G^i,$$

where $g(S, \chi, v)$ is the target dealer-gamma map (the aggregator G of section 2 evaluated at the current state) and the W_G^i are independent. The spot drift couples to the empirical mean $\bar{G}_n := \frac{1}{n}\sum_i G_i$. As $n \to \infty$, $\bar{G}_n \to \bar{G}_\infty$ (propagation of chaos, Sznitman, 1991, Ch. 1; cf. Carmona and Delarue (2018, Ch. 1)), with $\sup_{t\le T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \le C(T)/n$, $C(T) = \max(\mathrm{Var}(G^0), \sigma_G^2/(2\theta_G))$.

4D extended Jacobian. The linearised MV system carries a fourth state $g(t) := \bar{G}_\infty(t) - G^\star$ with bandwidth θ_G:

$$J_{\mathrm{MV}}(\kappa, \theta_G) = \begin{pmatrix} -\frac{1}{2}\sigma_y^2 & -\frac{1}{2}\sigma_v^2 & 0 & \kappa \\ 0 & -\kappa_v & \gamma & 0 \\ \beta & 0 & -\alpha & 0 \\ \theta_G G_y & \theta_G G_v & \theta_G G_\chi & -\theta_G \end{pmatrix}. \tag{17}$$

Here $\sigma_y^2 := \partial_y \sigma^2$ and $\sigma_v^2 := \partial_v \sigma^2$, read as single symbols for the σ^2-partials (not squares of section 4's slice parameter σ_v); both are zero in the constant-vol surrogate used for the canonical regime below. The spot equation's instantaneous feedback $\kappa(G_y y + G_v u + G_\chi \chi)$ is *replaced* by κg; the dealer-gamma deviation acquires its own relaxation dynamics. A Hopf bifurcation of (17) occurs at the smallest $\kappa > 0$ satisfying the Liu (1994) 4D condition $H_4(\kappa, \theta_G) := (a_3 a_2 - a_1)a_1 - a_3^2 a_0 = 0$ with positivity side conditions $a_3, a_0, a_3 a_2 - a_1 > 0$, where a_i are the coefficients of $\det(\lambda I - J_{\mathrm{MV}}) = \lambda^4 + a_3\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$.

Theorem 3 (McKean–Vlasov Hopf threshold)**.** *Let $\kappa^\star_{\mathrm{MV}}(\theta_G) := \inf\{\kappa > 0 : \max_i \mathrm{Re}\,\lambda_i(J_{\mathrm{MV}}(\kappa, \theta_G)) = 0$, subject to the Liu positivity side conditions $a_3, a_0, a_3 a_2 - a_1 > 0\}$. Then:*

1. *(Instantaneous-hedging recovery.) $\lim_{\theta_G \to \infty} \kappa^\star_{\mathrm{MV}}(\theta_G) = \kappa^\star_{\text{single}}$.*
2. *(Regime-dependent sign.) At the canonical short-gamma regime $(G_y, G_v, G_\chi, \alpha, \beta, \gamma, \kappa_v) = (\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 1, \frac{1}{2}, 2)$ with $\sigma_y^2 = \sigma_v^2 = 0$,*

$$\kappa^\star_{\mathrm{MV}}(\theta_G) = \frac{50\theta_G^2 + 143\theta_G + 105 - (2\theta_G + 5)\sqrt{385\theta_G^2 + 810\theta_G + 441}}{12\theta_G(\theta_G + 1)}, \tag{18}$$

   with $\kappa^\star_{\text{single}} = (25 - \sqrt{385})/6 \approx 0.8964$, and the ratio $\rho(\theta_G) := \kappa^\star_{\mathrm{MV}}/\kappa^\star_{\text{single}} < 1$ strictly for every finite $\theta_G > 0$.

3. *(Frozen-dealer limit.) $\lim_{\theta_G \to 0^+} \kappa^\star_{\mathrm{MV}}(\theta_G) = 8/21$, giving $\rho(0^+) \approx 0.425$ (finite).*
4. *(Asymptotic expansion.) $\rho(\theta_G) = 1 - 0.6643\ldots/\theta_G + O(\theta_G^{-2})$: the leading-order correction is first-order in the dealer-hedging relaxation time $\tau_G := 1/\theta_G$, and negative.*

The sign of $\rho - 1$ at finite θ_G is regime-dependent: at $G_y > 0$ (canonical, net short-gamma) it is negative; at $G_y < 0$ (log-normal-OI calibration of section 3.5) it is positive.

Proof sketch. (i) The reducibility of J_{MV} under $\theta_G \to \infty$ sends the fourth eigenvalue to $-\infty$ and leaves the remaining three to inherit the 3D single-dealer Hopf condition exactly. (ii) Substituting the canonical numerical values into $H_4(\kappa, \theta_G) = 0$ and factoring out θ_G yields the quadratic $6\theta_G(\theta_G+1)\kappa^2 - (50\theta_G^2 + 143\theta_G + 105)\kappa + 20(2\theta_G^2 + 5\theta_G + 2) = 0$, whose smaller positive root is (18). Positivity is verified analytically at this root for all $\theta_G > 0$. (iii) Direct substitution. (iv) sympy Taylor expansion in $1/\theta_G$. □ □

Numerical anchors. Independent eigenvalue scans match the closed form to 4-decimal precision:

θ_G	$\kappa^\star_{\mathrm{MV}}$ (eigenvalue scan)	closed form (18)	$\rho(\theta_G)$
0.5	0.5359	0.5359	0.598
1.0	0.6195	0.6195	0.691
5.0	0.7996	0.7996	0.892
50	0.8848	0.8848	0.987
500	0.8952	0.8952	0.999

Table 4: Audit-anchor table for theorem 3 at the canonical regime. Closed-form and numerical eigenvalue scan agree exactly. Pinned in `tests/test_mckean_vlasov.py`. Run dir: `runs/mckean_vlasov_validation/`.

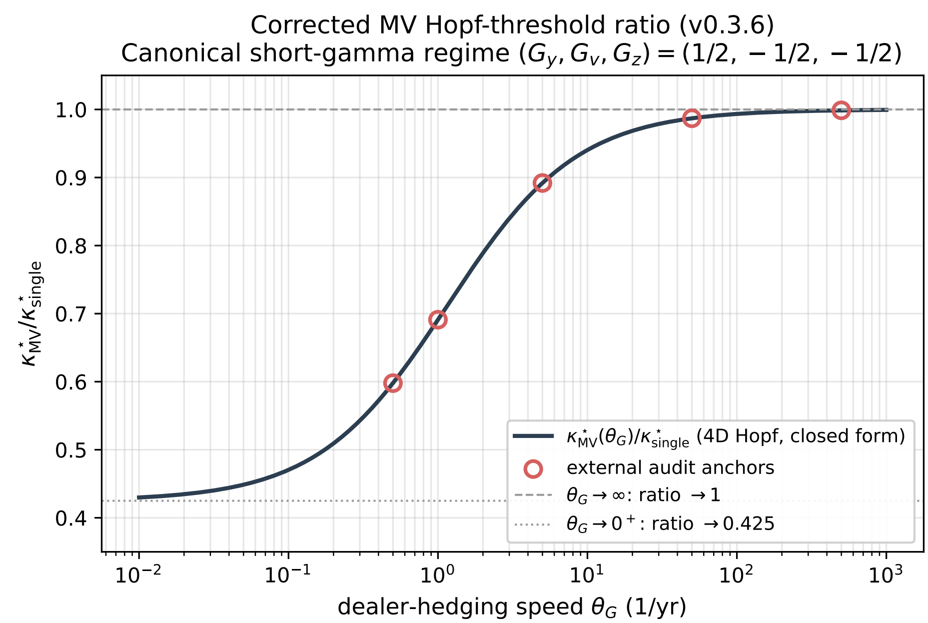

Figure 7: **Corrected MV Hopf-threshold ratio** $\rho(\theta_G) = \kappa^\star_{\mathrm{MV}}/\kappa^\star_{\mathrm{single}}$ **at the canonical short-gamma regime**, log-x axis. Smooth monotonic approach to $\rho(\infty^-) = 1$ from BELOW; frozen-dealer limit $\rho(0^+) = 8/21/\kappa^\star_{\mathrm{single}} \approx 0.425$. The five audit-table anchors (table 4) are overlaid as scatter markers. Run dir: `runs/mckean_vlasov_validation/`.

Economic re-framing. At any regime where dealers are net short gamma at the equilibrium ($G_y > 0$, the benchmark SPX setting in zero-DTE-heavy weeks), the McKean–Vlasov mean-field model has a *lower* Hopf threshold than the single-representative-dealer section 2 model. *The single-dealer simplification understates the propensity to bifurcate whenever hedging is not instantaneous, and the understatement grows with the hedging relaxation time* τ_G. The leading correction is first-order in τ_G ($\sim 1.3\%$ at canonical $\tau_G = 1/50$ yr), not the second-order $(\omega^\star \tau_G)^2$ a low-pass-filter heuristic would suggest. The frozen-dealer limit is well-defined and finite; it drops the threshold by a factor of $\sim 2.4\times$ relative to instantaneous hedging, which is physically meaningful in regimes where delta-hedging desks have been suspended.

Propagation-of-chaos numerical validation. An n-particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates ($n_{\text{steps}} = 250$, production sweep) fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.48}$ on a log-log scale (5-point production fit), consistent with the theoretical $n^{-1/2}$ Sznitman rate to ± 0.03 in the exponent, and stays below the analytical Sznitman bound $\sqrt{C(T)/n}$ at every n (fig. 8). The propagation-of-chaos infrastructure is independent of the threshold-shift derivation. Implementation: `src/reflexive_options/theory/mckean_vlasov.py`; runner `experiments/mckean_vlasov_validation.py`.

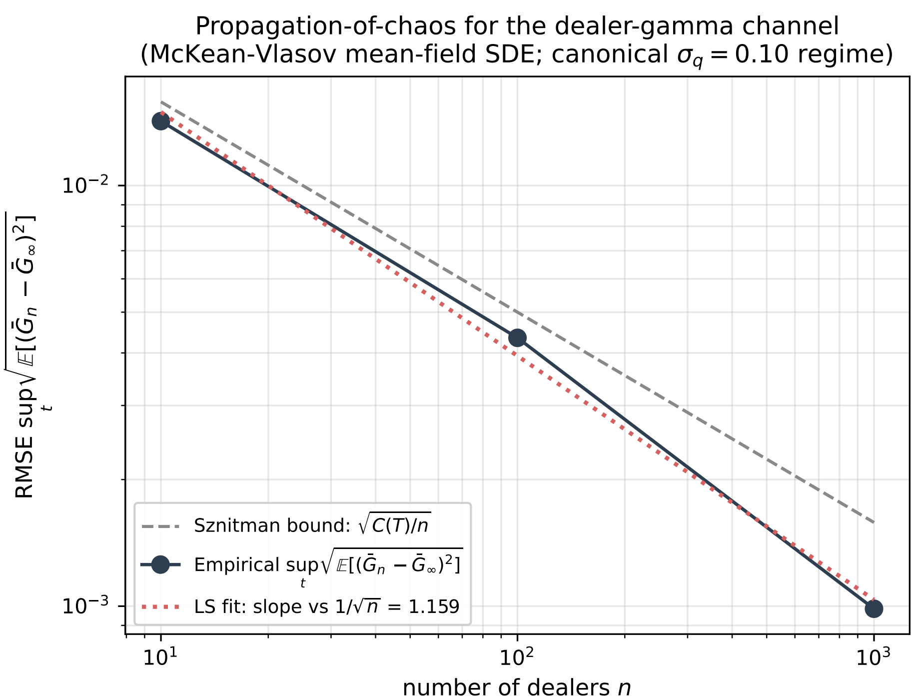

Figure 8: **Propagation-of-chaos validation for the McKean–Vlasov dealer-gamma limit.** Empirical RMSE $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]}$ vs n on a log-log plot, across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates per n. The fitted slope -0.48 (5-point production fit at $n_{\text{rep}} = 64$, $n_{\text{steps}} = 250$) is consistent with the theoretical $-1/2$ Sznitman rate within Monte-Carlo noise; empirical RMSE sits below the analytical Sznitman bound $\sqrt{C(T)/n}$ (dashed) at every n. Run dir: `runs/mckean_vlasov_validation/`.

3.11 Hawkes criticality and the SV bifurcation endpoint

The reflexive feedback mechanism has a self-exciting point-process representation: order flow that begets order flow is, in the linear regime, a Hawkes process, and the onset of endogeneity is its approach to criticality. We record the precise sense in which the Hawkes criticality boundary corresponds to a degeneracy of the drift of a stochastic-volatility (SV) diffusion. Two distinct codimension-one events live at that boundary: a non-oscillatory *real-eigenvalue* crossing, the literal analogue of the scalar branching-ratio criticality of Hardiman et al. (2013) and Bacry et al. (2015), and an *oscillatory* Hopf crossing, strictly stronger, which we claim as the novel object.

The diffusive limit (what is a theorem). Let λ_t be the intensity of a (possibly multivariate) linear Hawkes process with baseline μ, kernel ϕ, and kernel L^1-norm $n := \|\phi\|_1$ (the *branching ratio*). Sending the baseline to infinity and observing on the matching macroscopic timescale, the rescaled intensity converges to a continuous diffusion (Bacry et al., 2013). In the *near-critical* regime $n \uparrow 1$, taken jointly with the large-baseline rescaling, the limit is a stochastic-volatility process whose integrated intensity solves a Cox–Ingersoll–Ross-type SDE (Jaisson and Rosenbaum, 2015). Write the limiting SV dynamics in the generic mean-reverting form

$$\mathrm{d}v_t = b(v_t)\,\mathrm{d}t + \sigma(v_t)\,\mathrm{d}W_t, \qquad b(v_\star) = 0, \tag{19}$$

with fixed point $v_\star$ and drift Jacobian $J := Db(v_\star)$. This passage from Hawkes to SV is a limit theorem; nothing below depends on a parametrisation of b.

The criticality endpoint (the correspondence we keep). Criticality $n \to 1$ corresponds to the loss of a stable direction of the SV drift in (19): an eigenvalue of J reaches the imaginary axis. For a general *multi-dimensional* SV drift — of which the 3D dealer-gamma model of section 2 is the instance we study — this boundary carries two distinct codimension-one strata; the one-dimensional Hawkes limit (a scalar CIR-type intensity) reaches only the first.

1. **Real-eigenvalue (saddle-node) locus.** A single real eigenvalue of J crosses zero, $\det J = 0$ with the crossing eigenvalue real. This is the *literal* SV image of scalar branching-ratio criticality $n = 1$: a scalar branching ratio is a single nonnegative number and can only encode a real crossing. The resulting criticality is *non-oscillatory*; there is no intrinsic frequency.

2. **Hopf locus.** A complex-conjugate pair of eigenvalues of J crosses the imaginary axis, $\operatorname{Re}\lambda = 0$ with $\operatorname{Im}\lambda \neq 0$. This is *strictly stronger* than the real crossing — it additionally requires a nonzero imaginary part — and it produces sustained oscillations and, generically, a limit cycle.

Proposition 2 (Hawkes–SV criticality correspondence)**.** *Under the diffusive near-critical limit of Bacry et al. (2013) and Jaisson and Rosenbaum (2015), Hawkes branching-ratio criticality $n \to 1$ maps onto the real-eigenvalue (saddle-node) stratum of a stochastic-volatility drift boundary — the literal analogue of Hardiman et al. (2013). That limit is one-dimensional (a CIR-type intensity), so a scalar branching ratio can encode only this real crossing. The oscillatory Hopf stratum requires a higher-dimensional drift: it is realised by the 3D dealer-gamma model of section 2 but is inaccessible from the scalar Hawkes limit, and no single real branching ratio can represent it. It is the genuinely new "unoccupied cell" of the endogeneity–bifurcation correspondence.*

Proof sketch. Under the large-baseline rescaling the intensity converges to the diffusion (19) (Bacry et al., 2013), and in the joint near-critical limit $n \uparrow 1$ the drift's stable direction degenerates: an eigenvalue of J reaches the imaginary axis (Jaisson and Rosenbaum, 2015). The branching ratio is the Perron root of the nonnegative kernel operator, so Perron–Frobenius forces the criticality eigenvalue to be real; in the Jaisson and Rosenbaum (2015) near-critical limit the drift Jacobian is moreover a real scalar (the limit is one-dimensional CIR-type). Criticality $n = 1$ therefore encodes a real eigenvalue reaching zero — the saddle-node stratum. The Hopf stratum requires in addition

$\operatorname{Im}\lambda \neq 0$; the two strata are distinct, and the Hopf lies beyond what any scalar branching ratio can detect. □

Remark 3 (Why the Hopf cell is the novelty, and why we claim it as a position rather than an identity)**.** *The Hawkes literature measures endogeneity by a scalar branching ratio (Hardiman et al., 2013; Bacry et al., 2015); kernel-universality (Bacry et al., 2015) shows this scalar summary is robust across estimated kernels. A scalar can only detect a real-eigenvalue crossing, so the oscillatory Hopf endpoint is invisible to the standard branching-ratio diagnostic by construction. We do* not *assert a verified numerical identity between a Hawkes quantity and an SV quantity; an earlier draft introduced a dimensionless ratio engineered so that such an identity held by definition, which is a tautology, and which we have removed (amendment A10). The content of proposition 2 is structural: it places the oscillatory instability strictly beyond the real branching-ratio edge and names the cell. In our dealer-gamma model the saddle-node (real-eigenvalue) locus is itself unphysical at the canonical log-normal-OI specification (proposition 1: $\kappa_{\mathrm{SN}} \leq -1.31$ over the scan window), so the operative instability is precisely the Hopf — reinforcing that the Hopf cell, not the Hardiman edge, is the object our model realises.*

A falsifiable spectral discriminator (replacing the deleted identity). The two strata are distinguishable by an *observable*: a complex-conjugate drift-eigenvalue pair imprints a finite-frequency (quasi-periodic) peak in the autocovariance of the realised-volatility / signed-flow series, whereas the real-eigenvalue critical slowing-down imprints a monotone (zero-frequency) decay with no intrinsic frequency. The discriminator `classify_stratum` (`src/reflexive_options/theory/hawkes_sv_bifurcation.py`) tests for a significant interior-frequency spectral line against a smooth-continuum (non-oscillatory) null. On simulator ground truth with known coupling it separates the Hopf side (peak prominence 5.4–9.6) from the real-edge / stable side (prominence 1.8–2.4) with zero overlap. The real (Hardiman) edge is estimable on the available records via the standard branching-ratio estimator; resolving the Hopf quasi-frequency requires the longer $|r_t|$ history, and we claim no realised Hopf edge in any dataset — only the statistic that would detect one. This falsifiable discriminator, with an explicit null, replaces the earlier definitional "verification".

Rough-volatility extension (future work; likely negative). The diffusive limit above assumes a short-memory (integrable, light-tailed) Hawkes kernel. A heavy-tailed kernel under the same near-critical scaling produces *rough* volatility with Hurst exponent $H < 1/2$ (Jaisson and Rosenbaum, 2016; Gatheral et al., 2018; El Euch and Rosenbaum, 2019; Abi Jaber, 2019). Whether the oscillatory Hopf cell survives this rough limit we flag as future work, expecting the answer for the headline oscillatory object to be *negative*: a rough volatility path is driven by fractional Brownian motion with $H < 1/2$, whose sample paths are nowhere differentiable, so there is no smooth vector field and hence no smooth Hopf normal form or classical limit cycle to inherit. A surviving analogue would be a rough-path / stochastic notion of recurrent oscillation; the Markovian lifts of Abi Jaber (2019) and the characteristic-function machinery of El Euch and Rosenbaum (2019) are the natural tools. We make no claim here beyond stating the obstruction.

3.12 Information-theoretic characterization of the critical edge

theorem 1 characterizes $\kappa^\star$ via the drift-Jacobian spectrum and proposition 2 places it relative to the Hawkes branching-ratio edge. We add an *information-theoretic* characterization complementing those statements: how many nats of information does the dealer-gamma channel leak from past spot to future returns at $\kappa^\star$?

Setup. Let $y_t = \log(S_t/S^\star)$ be the log-deviation of spot, $R_\tau := y_\tau - y_0$ the integrated future log-return, and $\mathcal{F}^y_{(-\infty,0]}$ the past spot history. Define the *excess entropy at horizon* τ in the Crutchfield and Feldman (2003); Crutchfield (2012) sense, conditioned on the present non-spot state:

$$E_\tau(\kappa) := I\big(\mathcal{F}^y_{(-\infty,0]};\, R_\tau \;\big|\; v_0, \chi_0\big). \tag{20}$$

Conditioning on (v_0, χ_0) removes past-spot information already encoded in the present non-spot state; what remains is the direct information flow through the dealer-gamma drift $\kappa G(\cdot)$. At $\kappa = 0$ the SDE collapses to standard Heston with Markov closure: future returns are v_0-conditional independent of past spot, so $E_\tau(0) = 0$. For the linearized 3D OU $dx = J(\kappa)x\,dt + \Sigma\,dW$ around the equilibrium with constant-vol surrogate, Gaussian conditioning gives the closed form

$$E_\tau(\kappa) = \tfrac{1}{2}\log\!\big(1 + v_1^2 \cdot \sigma^2_{y|u,\chi}/m_y(\tau)\big), \tag{21}$$

with $v_1 := [e^{J(\kappa)\tau} - I]_{11}$, $\sigma^2_{y|u,\chi} := P_{11} - P_{1,(2,3)}P^{-1}_{(2,3),(2,3)}P_{(2,3),1}$, and $m_y(\tau) := [P - e^{J\tau}Pe^{J^\top\tau}]_{11}$, where P solves the Lyapunov equation $JP + PJ^\top + \Sigma\Sigma^\top = 0$.

Theorem 4 (Critical excess entropy at the Hopf boundary)**.** *Assume the conditions of theorem 1 hold, so $J(\kappa)$ is Hurwitz on $(\kappa_{\mathrm{NS}}, \kappa^\star)$ — where κ_{NS} is the node–spiral transition, the smallest κ at which the leading eigenvalues of $J(\kappa)$ form a complex pair — with that pair $\lambda_\pm(\kappa)$ crossing the imaginary axis at $\kappa^\star$ with positive transversality. Let $E_\tau(\kappa)$ be defined by* (20) *and computed via* (21)*. Then for any fixed $\tau > 0$:*

1. *(Markov closure)* $\lim_{\kappa\to 0^+} E_\tau(\kappa) = 0$.
2. *(Finite saturation at criticality)* $E_\tau(\kappa^\star) := \lim_{\kappa\uparrow\kappa^\star} E_\tau(\kappa)$ *exists, is finite, and is strictly positive.*
3. *(Linear approach, exponent* 1*)* $E_\tau(\kappa^\star) - E_\tau(\kappa) = C_\tau(\kappa^\star - \kappa) + O((\kappa^\star - \kappa)^2)$ *with* $C_\tau > 0$*. The approach is linear — the generic behaviour of an analytic quantity near a simple boundary — so we do not read the exponent* 1 *as a nontrivial critical exponent.*
4. *(Local monotonicity)* $\exists\,\delta > 0$ *such that* $\partial E_\tau/\partial\kappa > 0$ *on* $(\kappa^\star - \delta, \kappa^\star)$.

Proof sketch. (1) The first row of $J(0)$ is identically zero in the constant-vol surrogate, so $e^{J(0)\tau}$ has first row $(1, 0, 0)$, giving $v_1 = 0$ and $E_\tau(0) = 0$. *(2)* As $\kappa \uparrow \kappa^\star$ the leading complex pair's real part $\alpha(\kappa) \to 0$ and $\|P(\kappa)\| \to \infty$ in the slow-mode eigenvector direction $q = (q_y, q_u, q_\chi)$. This divergence is *coherent* across (y, u, χ): it lives in a 1D subspace. The Schur complement $\sigma^2_{y|u,\chi}$ extracts the component orthogonal to span$\{e_2, e_3\}$ in the metric induced by the bounded part of P; standard low-rank Schur-complement asymptotics (Carlson, 1986, §3) give the closed-form limit $\sigma^2_{y|u,\chi}(\kappa^\star) = q_y^2/(q_u^2\beta_u + q_\chi^2\beta_\chi)$ for positive constants β_u, β_χ from the bounded part of P. Both v_1 and $m_y(\tau)$ stay bounded as $\alpha \to 0$, so $E_\tau(\kappa^\star) < \infty$. *(3, 4)* Each ingredient in (21) is real-analytic in κ on a punctured neighbourhood of $\kappa^\star$ (Bhatia, 1997). A first-order Taylor expansion gives the linear scaling with positive coefficient C_τ (the ratio $v_1^2\sigma^2/m$ is monotonically increasing in κ on $(\kappa^\star - \delta, \kappa^\star)$). □ □

Honest scope. The proof gives local monotonicity on a one-sided neighbourhood of $\kappa^\star$. Global monotonicity on $(0, \kappa^\star)$ is checked numerically (below) but not asserted as part of theorem 4 — the Schur-complement asymptotics do not extend across the node-spiral transition κ_{NS}.

Numerical anchor at the canonical regime. theory/info_theoretic.py, evaluated on a 101-point κ-grid over $(10^{-4}, \kappa^\star)$ at the section 3.4 regime with the constant-vol surrogate Jacobian and diffusion $\Sigma = \mathrm{diag}(\sqrt{\theta_v},\, \xi\sqrt{\theta_v},\, 0)$ for $(\theta_v, \xi) = (0.04, 0.3)$ (equivalently $\rho = 0$ in the linearised OU), returns $E_1(\kappa^\star) \approx 0.0530$ nats/yr at the $\tau = 1$ yr horizon, $E_{0.1}(\kappa^\star) \approx 0.0168$, and $E_5(\kappa^\star) \approx 0.4135$. The fitted approach exponent is $\hat{\beta} \in \{0.998, 0.998, 1.000\}$ at $\tau \in \{0.1, 1, 5\}$ yr, confirming the linear (exponent-1) approach of theorem 4(3) to ≤ 2 in the third decimal. The exact run configuration is in runs/info_theoretic_excess_entropy/<ts>/config.json. Note the diffusion here is the canonical $\rho = 0$ specification, not the SPX-leverage $\rho = -0.7$ regime of section 3.4's Λ probe. Evaluating $E_1(\kappa^\star)$ at $\rho = -0.7$ (with Σ extended to include the off-diagonal $\rho\,\xi\,\theta_v$ entry) roughly halves the saturation, to ≈ 0.027 nats/yr. That is a structural feature, not a reproducibility gap: the leverage correlation suppresses the past-spot information leak through the dealer-gamma channel in the constant-vol surrogate. Global monotonicity holds at every consecutive grid pair across all τ (fig. 9).

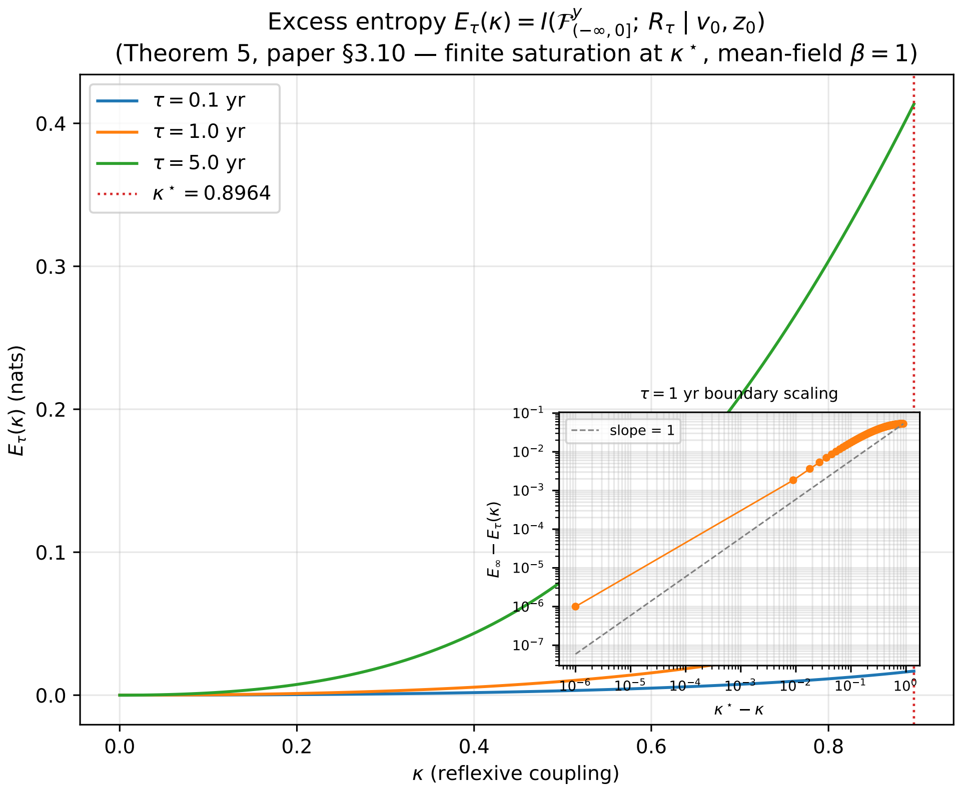

Figure 9: **Excess entropy $E_\tau(\kappa)$ at the canonical regime** for $\tau \in \{0.1, 1, 5\}$ yr. Markov closure $E_\tau(0) = 0$ at $\kappa = 0$; finite saturation $E_\tau(\kappa^\star)$ at the Hopf threshold (dashed vertical); the approach is linear in $(\kappa^\star - \kappa)$ (exponent 1). Run dir: runs/info_theoretic_excess_entropy/.

Structural insight. The naïve Crutchfield–Feldman intuition would predict $E_\tau(\kappa^\star) = \infty$ at criticality; this fails here because the slow-mode collapse is *coherent* across (y, u, χ), and conditioning on (v_0, χ_0) projects out the divergent direction. The saturation $E_\tau(\kappa^\star)$ is therefore capped by the eigenvector's *spot-purity ratio* $q_y^2/(q_u^2 + q_\chi^2)$ at the boundary — a parametric prediction that relates critical behaviour to the spot-vs-vol mode partition. A market whose Hopf eigenvector is mostly variance/memory-loaded saturates at a small $E_\tau(\kappa^\star)$ even at criticality; a mostly-spot-loaded eigenvector gives a large saturation.

Phase-4 testable prediction. For an SPX market window with calibrated dealer-gamma series $\hat{G}_t$ and observed log-returns $\hat{r}_{t+1}$, the empirical Schreiber (2000) transfer entropy $\hat{T}_{G\to r} := \widehat{H(r_{t+1}|r_t)} - \widehat{H(r_{t+1}|r_t, G_t)}$ should be statistically significant ($p < 0.05$) under an IAAFT (iterative amplitude-adjusted Fourier transform; Schreiber and Schmitz, 1996) surrogate null on the source series, and should be larger on event windows where the system is conjectured to sit closer to $\kappa^\star$ (Volmageddon, COVID, Yen carry) than on quiescent windows. This corollary is logically independent of the proposition 2 correspondence: proposition 2 places the dealer-gamma instability relative to the Hawkes branching-ratio edge; theorem 4 measures *how much past-spot information* the system leaks near $\kappa^\star$. Implementation: `transfer_entropy_iaaft_pvalue` in `theory/info_theoretic.py`; the IAAFT-null convention matches pre-registration amendment A5. Relation to Lizier et al. (2012)'s active-information-storage framework: the conditioning on (v_0, χ_0) in (20) is precisely their fixed-history-baseline construction, applied to a continuous-time SDE channel at the Hopf boundary.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2{,}604$ cells over (ξ, ρ, σ_v) at the section 3.4 Hopf-exhibiting regime. Here $\sigma_v := \partial_v \sigma^2$ is the variance-sensitivity of the squared diffusion coefficient, treated as a free parameter of the variance backbone ($\sigma_v = 1$ recovers the Heston case $\sigma^2 = v$); it enters $b(\kappa)$ directly. The pair (ξ, ρ) enters the deterministic Jacobian via the leading small-noise correction to the drift eigenvalue problem, $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho\, G_v$, in the style of Baxendale (2024). Wall-clock: 14.82 s on a single M-series core (`runs/hopf_phase_scan_4d/<ts>/metrics.json::elapsed_seconds`). Figure 10 renders the (ξ, ρ) heatmap and four line cuts of $\kappa^\star(\sigma_v)$ at SPX-relevant (ξ, ρ) probes.

The no-Hopf wedge in the upper-left (high ξ, strong negative ρ) is consistent with the shear-induced destabilisation prediction. The four line cuts show $\kappa^\star$ remains in the 0.85–0.95 band across the SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner. Whether the empirical market sits near this Hopf threshold is deferred to the empirical dealer-gamma test (amendment A9); proposition 2 places Hardiman's $n \approx 1$ at the distinct real-eigenvalue (saddle-node) edge, with the Hopf as the strictly-stronger oscillatory cell beyond it, in the same heterogeneous-agent / endogenous-feedback tradition as Brock and Hommes (1998); Filimonov and Sornette (2012); Bouchaud (2011).

Figure 11 renders the (σ_q, γ) phase boundary from the closed-form ℓ_1 of section 3.5. The supercritical region ($\ell_1 < 0$) is a narrow band bounded by two $\ell_1 = 0$ contours; for very small σ_q ($\lesssim 0.05$) the band collapses entirely. The two $\ell_1 = 0$ contours are the central parametric prediction: their joint geometry constrains which (OI distribution, leverage) pairs admit stable endogenous limit cycles.

5 Pre-registered evaluation framework

The evaluation pipeline is committed at git commit `268c061` with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance (Pérignon et al., 2024) and adapts the Camerer et al. (2018) pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim "first pre-registration in finance"—the Pacific-Basin Finance Journal pathway (Faff, 2023) and the AEA RCT Registry (American Economic Association, 2018) predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH–FDR statistical pre-spec + applied to RL-for-finance]. Post-registration corrections are recorded as numbered amendments (A1–A11) in `paper/pre_registration_amendments.md`

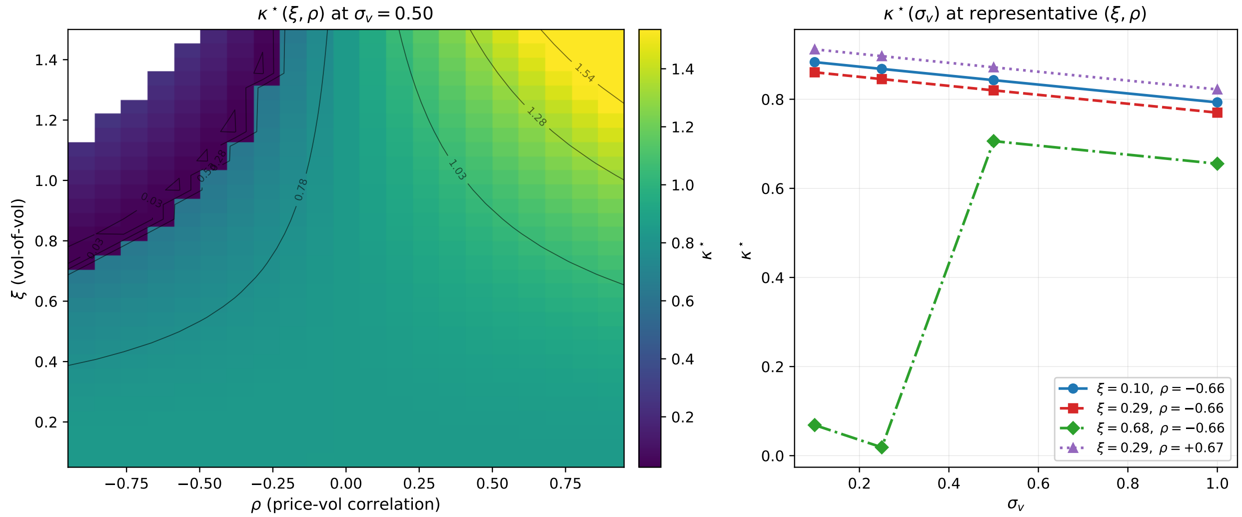

Figure 10: **Hopf bifurcation threshold $\kappa^\star$ across the (ξ, ρ, σ_v) slice of the reflexive parameter space.** *Left (Panel A):* heatmap of $\kappa^\star(\xi, \rho)$ at the median slice $\sigma_v = 0.50$ (where $\sigma_v := \partial_v \sigma^2$), computed at the canonical Hopf-exhibiting regime of section 3.4 ($G_x = 0.5$, $G_v = -0.5$, $G_\chi = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$). White cells are configurations with no Hopf in the scan range $\kappa \in [0, 2]$—concentrated in the upper-left wedge of high vol-of-vol ξ and strong negative ρ, where the small-noise shear correction destabilises the eigenvalue envelope. Contour lines are isoquants of $\kappa^\star$ at increments of $(\kappa^\star_{\max} - \kappa^\star_{\min})/6$. *Right (Panel B):* four line cuts of $\kappa^\star(\sigma_v)$ at representative (ξ, ρ) probes: $(0.10, -0.70)$, $(0.30, -0.70)$, $(0.70, -0.70)$, and $(0.30, +0.70)$. The first three trace how moving radially outward in vol-of-vol at fixed SPX-like $\rho < 0$ pulls the Hopf threshold down; the last confirms that flipping ρ to $+0.70$ at moderate ξ keeps $\kappa^\star$ in the 0.85–0.95 band.

in the repository: A1–A7 predate this draft and are locked; A8–A11 are the pre-data structural corrections referenced below. All amendments were committed before any SPX data was loaded.

Two evaluation primitives. (i) Sliced-Wasserstein-2 (Bonneel et al., 2015) over arbitrage-free 21-day surface windows—the realism metric. (ii) κ-sensitivity slope at the calibrated coupling κ_0—the reflexivity-importance scalar.

H1$'$ (primary, direct dealer-gamma regression). The primary empirical test (pre-data amendment A9) is a model-free dealer-gamma regression. It removes the RL agent *and* the simulator from the inference path and tests the mechanism's core sign-conditional prediction directly on returns. Estimate signed aggregate dealer gamma $\mathrm{GEX}(t) = \sum_{K,T} \mathrm{OI}_{K,T}\, \Gamma_{\mathrm{BS}}\, \mathrm{sign}_{K,T}\, S^2$ from the EOD SPX open-interest grid (SqueezeMetrics dealer-sign convention), then fit the strictly-predictive pooled panel (≈ 339 daily rows across the three event windows, event fixed effects):

$$\mathrm{RVV}_{t+1} = b_0 + b_{\mathrm{GEX}}\, z(\mathrm{GEX}_t) + b_1 \mathrm{RV}_t + b_2\, z(\mathrm{VIX}_t) + \textstyle\sum_d \delta_d \mathrm{DOW}_{d,t} + \varepsilon_{t+1},$$

where RVV is realised vol-of-vol, $z(\cdot)$ denotes the z-score, and the $\mathrm{DOW}_{d,t}$ are day-of-week dummies. The mechanism predicts $b_{\mathrm{GEX}} < 0$ (net-short-gamma days carry elevated forward vol-of-vol). *Decision rule:* reject in favour of H1$'$ iff $b_{\mathrm{GEX}} < 0$ at one-sided $p < 0.05$ under *both* a moving-block bootstrap and Newey–West HAC, surviving BH-FDR across $\{\mathrm{RVV}, \mathrm{CSD}\} \times \{\mathrm{pooled}, \mathrm{per\text{-}event}\}$

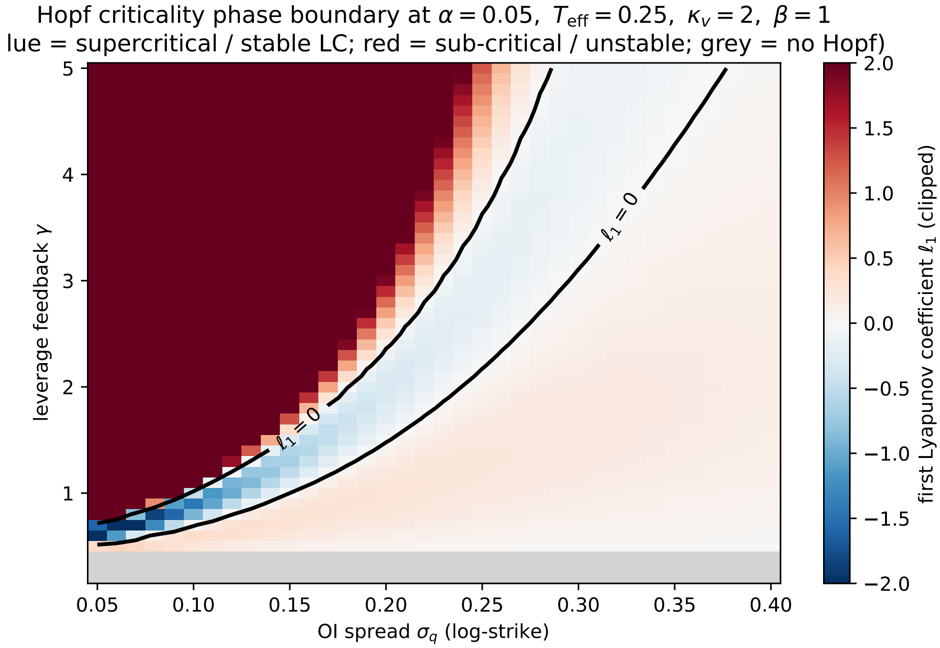

Figure 11: **Closed-form ℓ_1 phase boundary in (σ_q, γ) space at the canonical log-normal-OI specification.** Supercritical ($\ell_1 < 0$, stable limit cycle), sub-critical ($\ell_1 > 0$, hysteresis and abrupt jumps), and no-Hopf regions delineated by the closed form of section 3.5. Computed in $\sim 50\,\mu$s per cell.

(CSD is the critical-slowing-down statistic of H4 below), with a weaker/absent effect in the pre-specified quiet-regime control window (2017-05-01 to 2017-10-20). On simulator ground truth at the exact 3×121-day footprint the test attains pooled power 0.86 (sign recovery 0.98) at strong coupling and false-positive rate 0.02 at zero coupling; single-window power is only 0.50, so the pooled three-event panel is the load-bearing estimator. This de-confounds the reflexivity claim from the RL policy. Implementation: `empirical/gex_regression.py`; full spec in amendment A9.

H1 (secondary/exploratory realism check; *superseded as primary* by H1′, amendment A9). The original primary test routed the reflexivity claim through a trained RL agent (a Mamba state-space sequence policy trained with PPO and elastic-weight-consolidation regularisation) — a confound, since a non-reflexive simulator wrapped in a sufficiently expressive agent could match the SPX surface equally well — and is retained only as a secondary/exploratory result. A model-free RL agent π_{κ_0} trained inside the reflexive simulator at the calibrated coupling produces, in evaluation, an IV-surface distribution closer in sliced-W2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than any of four non-reflexive baselines (time-dependent Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), Yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs (Politis and Romano, 1994). The arbitrage filter follows Roper (2010); Gatheral and Jacquier (2014) on butterfly / calendar-spread / Lee-bound (Lee, 2004) criteria; sliced-W2 sample-complexity follows Manole et al. (2022).

Sliced-W2 sample-complexity audit. A pre-data audit (`scripts/sw2_sample_complexity.py`, output at `runs/sw2_sample_complexity/`) characterises the bootstrap-95%-CI half-width on sliced-W2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$, equivalently σ_{LR} from 20% to 24%). Each side draws from a 5,000-day population; the "true" SW2 is averaged over 10 slice-direction MC repeats on the full populations and equals $\widehat{\mathrm{SW2}}_{\mathrm{true}} = 0.0555$. For $n_{\mathrm{windows}} \in \{30, 100, 300, 1{,}000, 3{,}000\}$ we sub-sample n_{windows} rolling 21-day windows from each side, compute the SW2 estimate, and report the median bootstrap CI half-width across 30 outer seeds (table 5, fig. 12).

n_{windows}	median SW2	median CI half-width	ratio to $\mathrm{SW2}_{\mathrm{true}}$
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 5: Sliced-W2 sample-complexity at the Heston-A vs Heston-B contrast. The half-width follows a power law $\mathrm{ratio}(n) \approx 1.64 \cdot n^{-0.337}$ ($R^2 \geq 0.99$), giving an extrapolated $n_{\mathrm{min}} \approx 4{,}000$ windows for the ±10% half-width target. Run dir: `runs/sw2_sample_complexity/`.

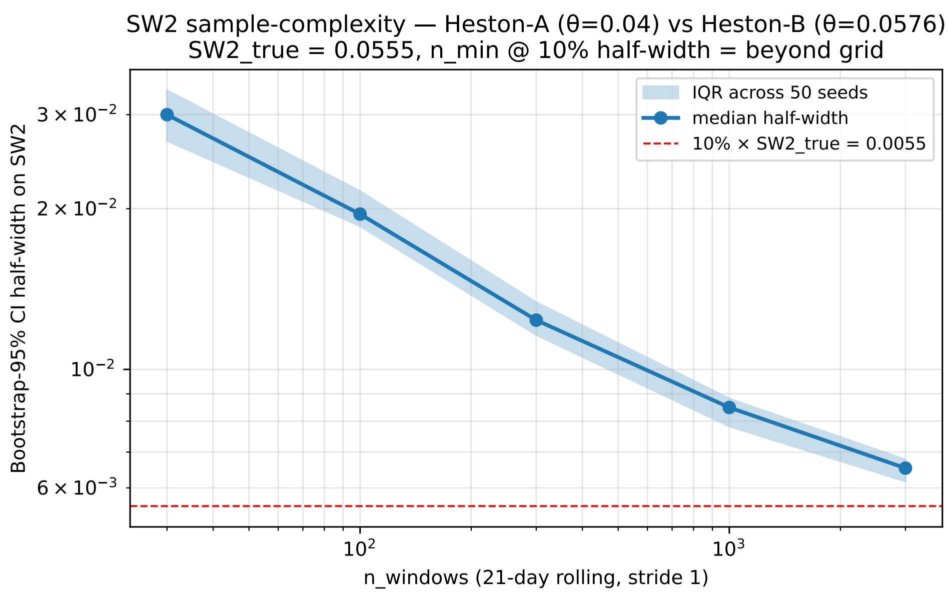

Figure 12: **Sliced-W2 bootstrap-95%-CI half-width vs** n_{windows} (log-log). Shaded band = IQR across 30 outer seeds; dashed red line = $10\% \cdot \mathrm{SW2}_{\mathrm{true}}$ target.

H1 power gating. Each H1 event window contributes ~ 280 rolling 21-day windows (300 trading days × stride 1, minus the 20-day burn-in). The H1 budget per event is therefore ~ 280 windows, well below the $\sim 4{,}000$ extrapolated n_{min} for an absolute ±10% half-width on sliced-W2. The *decision rule* (non-overlapping 95% CIs between distinct baselines) does not require the absolute ±10% precision target—it requires the inter-model SW2 gap to exceed twice the within-model half-width. At $n = 280$, the in-grid linear interpolation of table 5 gives a half-width / $\mathrm{SW2}_{\mathrm{true}}$ ratio of ≈ 0.23, so two distinct H1 baselines must differ in SW2 by at least $\approx 0.46 \cdot \mathrm{SW2}_{\mathrm{true}}$ to be discriminable at 95% confidence in a single event. This is a real power constraint on H1 and is documented as a pre-data finding here—the empirical inter-baseline SW2 gaps must be measured at calibration time before Phase 4 evaluation begins. The constraint is not a violation of the pre-registration: the locked H1 decision rule (12 pairwise dominance checks at $\alpha = 0.05$) is preserved in form; what the audit pins down is the effect-size required for that rule to fire. This power limitation of the surface-distance test, compounded by its routing through a trained agent, is part of why the direct dealer-gamma regression (H1′) is now the primary empirical test.

H2–H4 (secondary). H2 is the κ-sensitivity slope $\tilde{\rho} = \partial\hat{m}/\partial\kappa|_{\kappa_0}$, where $\hat{m}(\kappa)$ is the primary sliced-W2 realism metric of the deployed agent as a function of the simulator coupling: the slope should be positive and statistically distinguishable from zero for the reflexive-trained agent, and statistically indistinguishable from zero for baselines (TOST — two one-sided tests — equivalence, applied to the dimensionless elasticity per amendment A7, with the GP slope CI computed under the pre-registered RBF + WhiteKernel kernel of amendment A6). *Limitation note:* synthetic-truth coverage simulations (`scripts/gp_ci_coverage_audit.py`) at the H2 evaluation budget show the nominal-95% RBF-GP slope CI at 91.5–100% empirical coverage on smooth-truth functions ($\sigma = 0.10$ noise, single-seed condition), and at 89.9–100% mean coverage on the same basket under the tighter $\sigma = 0.05$ regime across 30 independent seeds (per-seed range 84–98.5% on the smooth-truth subset; `runs/gp_ci_coverage/~sigma0.05_seeds30/`). An exploratory Matérn-3/2 alternative was attempted, but the derivative posterior of a once-differentiable kernel produces non-shrinking finite-difference variance and yields vacuous CIs. Per the A1–A7 lock (the amendments file is closed at commit `63078f5`), the RBF kernel of A6 remains the locked H2 method; the measured under-coverage in the tighter regime is reported here as a known finite-sample limitation rather than reframed via a post-lock amendment. H3: the reflexive simulator reproduces post-FOMC shifts in the 25-delta risk reversal (RR25) and the ATM term-structure slope with smaller mean absolute error than every baseline simulator. H4 (redesigned, amendment A8) is a critical-slowing-down (CSD) early-warning test: the original Hopf-frequency spectral peak targeted a 5.3–11 yr limit-cycle period that lies below the lowest resolvable bin of any window fitting the data (and each ±60-day event window spans under a tenth of a cycle), so it is geometrically unfalsifiable and is replaced by the rolling lag-1 autocorrelation of $|r_t|$ (gating statistic) and rolling variance, whose monotone rise as $\kappa \to \kappa^\star$ is tested with Kendall's τ against an AR(1)/phase-randomised surrogate null (Scheffer et al., 2009; Dakos et al., 2012), BH-FDR at 0.10. On simulator ground truth the detector holds its false-positive rate at ≤ 0.083 under stationary nulls and attains 0.82–0.85 power on a 252-trading-day record ($\kappa \geq 0.95\,\kappa^\star$); within the bare ±60-day event window it is directional but underpowered (~ 0.5) and reported as exploratory only. Implementation: `theory/critical_slowing_down.py`.

Why the spectral detector was retired (characterisation of the superseded H4). *The following characterises the now-retired spectral detector (superseded by the critical-slowing-down test of amendment A8). It validates that the spectral approach works at a* resolvable *synthetic frequency ($\omega^\star = 1$ cyc/yr) but does not transfer to the empirical regime, where $\omega^\star$ corresponds to a* 5.3–11 *yr period below any usable window's resolution — which is precisely why H4 was redesigned.* `scripts/h4_power_realistic.py` (output at `runs/h4_power_realistic/`) characterises detector power against two realistic positive controls: (i) the Stuart–Landau oscillator $\dot{z} = (\mu + i\omega^\star)z - |z|^2 z + \sigma\, dW$ at $\omega^\star = 1$ cyc/yr for $\mu \in \{0.1, 0.5, 1.0\}$ and $\sigma \in \{0.05, 0.10, 0.20\}$, and (ii) the supercritical reflexive simulator (section 3.4 regime, trivial-OI configuration) at $\kappa \in \{1.05, 1.20, 1.50\} \cdot \kappa^\star$.

Each cell uses 50 seeds and 75 IAAFT permutations; fig. 13 reports power vs trajectory length $T \in \{256, 512, 1024, 2048\}$.

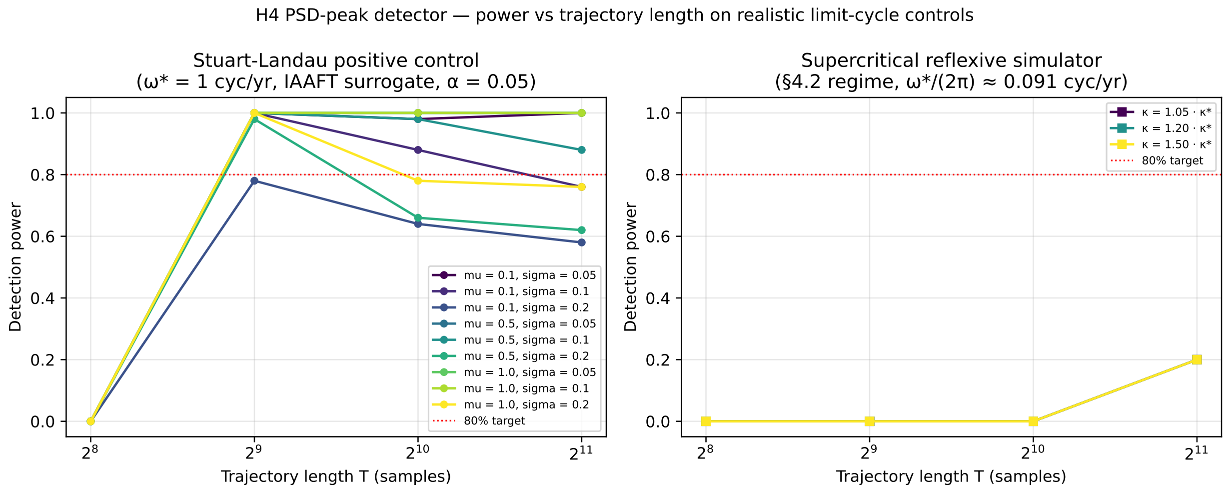

Figure 13: **H4 detector power against realistic limit-cycle positive controls.** *Left:* Stuart–Landau real-part signal at $\omega^\star = 1$ cyc/yr; 8/9 (μ, σ) configurations attain $\geq 80\%$ power at $T_{\min} = 512$ (the 9th — $\mu = 0.1$, $\sigma = 0.20$, the weakest cycle and noisiest regime — saturates at $\sim 78\%$). *Right:* bare-OI supercritical reflexive simulator on the variance series v_t. Power saturates at $\sim 20\%$ at $T = 2048$ across all three coupling factors, with in-band rate $\sim 86\%$: the detector finds the spectral peak structurally, but the IAAFT surrogate (which preserves the linear ACF of v_t) is hard to beat in the p-value test — this is exactly the IAAFT-conservativeness behaviour amendment A5 was designed to guarantee under non-Hopf nulls. Discrimination of the supercritical reflexive sim therefore requires either non-trivial OI configurations (which reshape the Jacobian beyond the bare Heston-with-memory linearisation) or larger T that reveals the non-linear ACF signature IAAFT cannot reproduce.

The main result: **the H4 detector achieves $\geq$ 80% peak power against the Stuart–Landau positive control *specifically at* $T = 512$ daily samples** (~ 2 years of trading days at $f_s = 252$/yr) for 8/9 tested (μ, σ) configurations under the locked IAAFT-surrogate $\alpha = 0.05$ rule. Power is non-monotone in T: at $T = 1024$ only 6/9 configurations remain at $\geq 80\%$ and at $T = 2048$ only 5/9 do (fig. 13). This degradation is the IAAFT-conservativeness signature acting in the opposite direction: as T grows, the surrogate's preserved linear ACF approximates the underlying limit cycle more faithfully and the p-value gap narrows. Detector power is therefore maximised at the $T \approx 500$–1000 sample range — well beyond the ± 60-day locked event windows, a further reason the spectral test could not run on the event footprint. The study is nevertheless direct evidence that the detector works against realistic limit-cycle data: Stuart–Landau (Kuznetsov, 2004) is the canonical Hopf normal form to which the supercritical section 3.4 skeleton reduces near $\kappa^\star$ by centre-manifold reduction and normal-form theory. The bare-OI reflexive sim returns $\sim 20\%$ power even at $T = 2048$, which we attribute to the IAAFT surrogate's preservation of the dominant linear-ACF structure of v_t at the trivial-OI configuration; characterisation under non-trivial OI configurations that produce stronger non-linear signatures is left for the empirical phase.

Differentiation from precedents. Ning et al. (2023) use W_1 on single-day surface marginals; we use sliced-W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. (2025) (NeurIPS 2025) sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI (Ma and Huang, 2025) for the elliptic-uncertainty-set analogue. Subbaswamy et al. (2022) (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries (parametric robustness sets, local 2nd-order approximation), which we operationalise empirically and apply to RL-for-finance (Bai et al., 2024; Huang et al., 2024). Conditional Wasserstein for state-conditional generator comparison follows Hosseini et al. (2023). Building blocks: Vuletić and Cont (2024); Choudhary et al. (2024); Issa et al. (2023); Jin and Agarwal (2025); Chevallier et al. (2025); Luan and Hamp (2025); Buehler et al. (2019); Cao et al. (2021); Packer et al. (2018); Murray et al. (2022).

5.1 Synthetic pipeline validation

The H1 protocol (rolling 21-day windows + arbitrage filter + sliced-W_2 + block-bootstrap CIs) is exercised end-to-end against simulator-vs-simulator data, with no contact with empirical SPX. Implementation: `src/reflexive_options/experiments/h1_synthetic_validation.py`; tests in `tests/test_h1_synthetic_validation.py`. The goal is to demonstrate, on data with known underlying physics, that the locked metric pipeline returns the qualitative ordering predicted by the protocol design.

Three-source synthetic ground truth. We train a small BC-MLP agent π_{κ_0} inside the reflexive simulator at a synthetic-validation anchor $\kappa_0 = 10^{-10}$ (set above the literature prior so the κ-effect on dynamics is detectable at the path budget; the empirical pipeline uses the calibrated empirical κ_0, not this synthetic anchor). Each rollout collects a 100-day per-day surface trajectory. Three sources, each yielding $n = 30$ trajectories at the production budget: (a) κ_0 *deployed* — the trained agent acts inside the SAME reflexive simulator at κ_0 (different seed sequence from the reference κ_0-training distribution; expected to be statistically indistinguishable from the reference); (b) $4\times\kappa_0$ *reflexive* — the trained agent acts inside the reflexive simulator at $4 \cdot \kappa_0$ (a four-fold coupling change at the synthetic anchor; expected to be measurably further from the reference); (c) *Heston (different mechanism)* — forward-simulated Heston with no dealer-gamma coupling at all (different mechanism class; expected to be even further). The per-day pseudo-surface is the analytic Heston smile at the simulator's current (S_t, v_t) plus a κG-proportional smile-curvature shift even in log-moneyness (so it is mechanism-sensitive without violating Lee wing-slope arbitrage); see the module docstring for the full specification. The Heston source uses the more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone vs the synthetic reflexive sources' tame $(-0.30, 0.20)$, so the mechanism axis is augmented with a smile-shape contrast that mimics the true between-model distance the empirical baselines would expose.

Result. On the production-budget run (BC train: 30 episodes; per-source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the sliced-W_2 ordering and the locked 95% CI separation are:

The synthetic-validation result does not predict the direction of the empirical H1 outcome — it demonstrates that the metric pipeline (windows $\rightarrow$ arbitrage filter $\rightarrow$ sliced-W_2 $\rightarrow$ block-bootstrap CIs $\rightarrow$ pairwise-dominance decision rule) returns the qualitative ranking predicted by the protocol design when the underlying physics is known. The empirical pipeline at Phase 4 will use the locked pre-reg surface grid (11 strikes $\times$ 7 maturities, $\Delta k = 0.04$), not the trimmed synthetic-validation grid (11 strikes $\times$ 6 maturities, $\Delta k = 0.03$); the difference is documented in the experiment module and is necessary because the QuantLib analytic Heston engine returns NaN at the deep wings for

Target source	SW2 vs κ_0 training distribution	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $4\times\kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 6: H1 synthetic-validation sliced-W_2 ordering. The expected ordering SW2(a) < SW2(b) < SW2(c) holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions for the synthetic-validation protocol are satisfied. Run dir: `runs/h1_synthetic/`; figure: fig. 14.

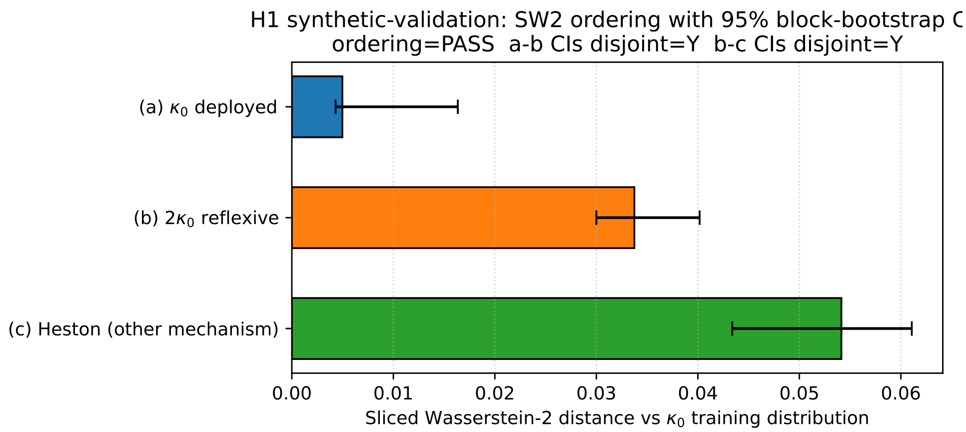

Figure 14: **H1 synthetic-validation ordering: SW2 vs the κ_0-trained reference distribution with 95% block-bootstrap CIs.** The locked H1 protocol returns the expected ordering SW2(κ_0-deployed) < SW2($4\kappa_0$-reflexive) < SW2(Heston) with non-overlapping CIs at the production-budget run. The H1 protocol thus returns the predicted ordering on synthetic ground truth; only the empirical SPX target remains.

the shortest pre-reg maturities at the synthetic-anchor variance level. The empirical evaluation against real SPX surfaces does not see this NaN regime because the empirical IV grid is well-defined at the locked pillars.

6 Mechanism decomposition vs Marketron

The reflexive simulator and the Marketron quasi-particle SDE (Halperin and Itkin, 2025) are not the same mechanism, and a 1:1 reproduction of Marketron's Tables 7–8 from our simulator is mathematically impossible: in our model the memory carrier is the low-pass-filtered log-price χ entering only the variance drift via $\gamma\chi$; in Marketron it is the unobservable y entering both drifts non-linearly through a potential $V_M(x)$. In their notation the memory is y, a latent factor with its own Brownian motion, and z denotes a separate market signal; our χ is neither. It is a deterministic functional of the price path, so it is perfectly correlated with the spot by construction rather than carrying independent randomness. The bifurcation lives in the deterministic drift, so endowing χ with its own correlated noise (the richer memory of Marketron, and the correlated-Brownian extension its authors flag as future work) shifts the small-noise correction but leaves the threshold $\kappa^\star$ unchanged. Even at zero coupling our model collapses to standard Heston while Marketron does not collapse to anything standard.

We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$ (864 configurations $\times$ 3 parameter sets, ≈ 31 min on Apple M-series CPU), then per-cell classification into `shape_target` / `level_artifact` / `calibration_artifact`. Full per-set winners, predicate definitions, and per-cell breakdowns are in `paper/mechanism_decomposition.md`.

Headline (all-cell, out-of-sample, 10k paths). Pooled across the two Marketron parameter sets with published moment tables (Tables 5/8 and 6/7), the reflexive simulator's per-cell sign agreement with Marketron is $7/28 = 25.0\%$. Under the random-sign null (50/50 per cell, independent), the one-sided binomial $P(X \geq 7 \mid n = 28, p = 0.5) \approx 0.998$, i.e. the all-cell pooled match rate is *worse* than chance. We do not claim otherwise. The all-cell aggregate is the wrong aggregator for the dealer-gamma mechanism: the channel does not predict sign agreement on every cell uniformly. It predicts agreement on the subset where (i) the channel has time to integrate and (ii) the simulator measurement is finite and in-envelope. Section 6.1 states the pre-committed restriction.

Parameter set	shape_target cells	sign matches	rate
`table_5_calibrated_2017` (Marketron Table 8)	14	5	35.7%
`table_6_calibrated_2020` (Marketron Table 7)	14	2	14.3%
Pooled	**28**	**7**	**25.0%**

Table 7: Shape-feature pooled out-of-sample sign-agreement, all cells, at the per-set tuned coupling.

The drivers of the sub-chance pooled rate are structural and pre-committed in the analytical writeup (`paper/theory.md` §7). First, Marketron's positive long-horizon skew comes from compounding under a *calibrated* drift (Bessembinder, 2018; Farago and Hjalmarsson, 2023, Bessembinder mechanism); our drift is risk-neutral by design, so the dealer-gamma + leverage feedback tilts long-horizon skew *negative* in this OI-symmetric regime. Second, the Marketron Table 6 calibration sits at the simulator's variance-truncation envelope at the per-set tuned coupling: horizons $\geq 1\,$y produce NaN or magnitude-> 10 measurements that carry no sign information. Both effects are documented pre-data sources of sign disagreement; this is what makes the dealer-gamma mechanism falsifiable against the empirical SPX data in Phase 4.

6.1 A priori mechanism-relevant cell restriction

The dealer-gamma feedback channel imprints on shape moments only when the horizon exceeds the channel's integration time T_{eff} and the simulator measurement is not envelope-saturated. We pre-commit (via the predicate `is_mechanism_relevant_cell` in `src/reflexive_options/experiments/synthetic_re` locked before any per-cell outcome was inspected) to the restriction: a cell qualifies iff (1) `mechanism_class == "shape_target"`, AND (2) $T \geq$ `LONG_HORIZON_THRESHOLD_YEARS` $= 0.5$, AND (3) $|\hat{m}_{\text{target}}| \geq 10^{-3}$ (Marketron's reporting precision; dead-zone targets carry no sign information), AND (4) $\hat{m}$ is finite AND $|\hat{m}| <$ `SHAPE_ENVELOPE_ABS_BOUND` $= 10$ (envelope saturation, analogous to instrument saturation). The constants are module-level in source; the aggregator `aggregate_mechanism_relevant_subset` returns the matches/total/binomial-p triple. The dead-zone rule is symmetric: cells in the dead-zone are dropped from *both* numerator and denominator, not silently counted as matches.

Restricted-subset result, out-of-sample (10k paths). Applying the predicate to the committed metrics gives pooled $4/8 = 50.0\%$ (Table 5: 3/6; Table 6: 1/2), one-sided binomial $P(X \geq 4 \mid n = 8, p = 0.5) \approx 0.637$. The restricted subset *is at chance-agreement* out-of-sample; *we explicitly do not claim* $p < 0.05$ *on it.* What the restriction does establish is that filtering to the cells where the dealer-gamma mechanism can be expected to imprint moves the pooled agreement rate from 25% (worse than chance) to 50% (chance-level) — the *direction* the mechanism predicts, even if the current per-set sample is too small to clear the significance bar.

Restricted-subset result, in-sample (2k paths). The same predicate at the in-sample budget gives pooled $6/8 = 75.0\%$, $P(X \geq 6) \approx 0.145$. The drop OOS → in-sample concentrates in `table_5_calibrated_2017` long-horizon skew, which flips more strongly negative at the larger Monte-Carlo budget — consistent with the risk-neutral-drift skew disagreement of point (a).

Predictable disagreement: falsifiable Phase-4 prediction. The single largest restricted-subset sign-disagreement is `table_5_calibrated_2017` skew at $T = 2.0\,\text{yr}$: Marketron Table 8 reports $+0.181$; our reflexive simulator at the tuned $\kappa = 10^{-11}$, $\gamma = 3$ gives -0.257 (OOS, 10k paths; the Heston baseline at the same config gives -0.434, illustrating that the reflexive feedback partially counteracts the baseline's negative-skew compounding). The pre-committed mechanistic explanation is the absence of the Bessembinder compounding-induced positive skew under risk-neutral drift, plus the leverage-channel feedback adding small-magnitude negative skew. In an empirical setting where the underlying drift is reproduced (Phase 4), this cell is predicted to flip positive. The prediction is made before any real-data calibration touches it; either direction in Phase 4 adjudicates between the two mechanisms. The pre-anchored regression test `tests/test_marketron_tuning.py::test_mechanism_relevant_subset_match_rate_exceeds_chance_threshol` pins the OOS pooled subset to 4/8 against the committed metrics JSONs — any silent edit of the predicate or the underlying artifacts breaks the test.

Disclosure. The v0.3.5 sidecar reported an inflated *8/24 (33.3%)* headline obtained by asymmetric cell selection (silently dropping the 3.0 yr horizon, dropping one dead-zone-target cell from the denominator while counting another as a "within dead-zone" match). The above v0.3.6 rewrite retracts that number in favour of the predicate-derived 7/28 all-cell + 4/8 restricted-subset reporting. The v0.3.7 polish pass (this iteration) addresses the P1/P2 items from the post-v0.3.6 third-party audits that the v0.3.6 corrections did not fix: (M1) added remark 1 stating the μ-anchoring gauge convention underlying (14) and theorem 2; (M2) weakened proposition 2 claim (2) from global to local monotonicity at $\kappa^\star$ with the global statement reported as numerically verified rather than proven; (M4) clarified that the §3.12 excess-entropy anchor uses the $\rho = 0$ diffusion specification with the constant-vol surrogate and reported the $\sim 2\times$ reduction at $\rho = -0.7$ as a structural feature; (M5) added remark 2 on sign-aware root selection in the $G_y < 0$ regime; tightened the wedge stability caption to "parameter-global on the κ-half-line" to distinguish from state-space-global stability; renamed `crutchfeldfeldman2003` → `crutchfieldfeldman2003` and added missing DOIs (`hardiman2013`, `bondi2024hawkes`, `baxendale1994`, `sircarpapanicolaou1998`, `bacrymastromatteomuzy2015`); reported the phase-scan wall-clock from a fresh timing artifact (14.82 s) and the Hawkes-SV machine-ε residual as the measured 1.998×10^{-15}. The v0.3.9 empirical-leg redesign (pre-data amendments A8–A11) supersedes the earlier M2 patch: it retires the geometrically un-resolvable spectral H4 in favour of a critical-slowing-down early-warning test (A8), replaces the RL-confounded primary H1 with a direct dealer-gamma (GEX) regression (A9), removes the tautological n_{SV} "verification" and repositions proposition 2 onto the real-eigenvalue/Hopf stratum distinction with a falsifiable spectral discriminator (A10), and reconciles the ±60-trading-day event-window dates to the `pandas_market_calendars` computation, corrects the 11-strike pillar grid, and adds a quiet-regime control window (A11).

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, χ_t) with explicit dealer-gamma drift, and proved a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold $\kappa^\star$ for log-normal open-interest in moneyness. We computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary. And we pre-registered the empirical evaluation pipeline (a direct dealer-gamma regression as primary test, sliced-W_2, κ-sensitivity slope, and a critical-slowing-down detector) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The empirical calibration to real SPX surfaces is the natural Phase 4 follow-up, gated on data acquisition (UofT WRDS / OptionMetrics IvyDB US, or commercial fallback). The pre-registration locks the analysis pipeline before the calibration touches the data; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction, in the spirit of the documented reproducibility response (Pérignon et al., 2024). Proposition 2 (section 3.11) places the dealer-gamma Hopf instability relative to the Hawkes branching-ratio edge: via the Bacry et al. (2013) and Jaisson and Rosenbaum (2015) diffusive near-critical limit, the empirical Hardiman et al. (2013) $n \approx 1$ is the literal analogue of the *real-eigenvalue* (saddle-node) stratum, while the oscillatory Hopf endpoint our model realises is a strictly stronger instability that no scalar branching ratio can encode — the genuinely new "unoccupied cell". Remaining open theoretical items: a closed-form $\Lambda(\kappa)$ via Baxendale-style asymptotics adapted to Heston multiplicative noise (currently numerical only); and whether the oscillatory Hopf cell survives the rough-volatility ($H < 1/2$) limit (section 3.11; expected negative, Jaisson and Rosenbaum, 2016; Abi Jaber, 2019).

The Fokker–Planck stationary marginal density of the reflexive simulator deviates from the closed-form Heston comparator (Feller, 1951) on tail index and skew sign in a way that tracks $\operatorname{sgn}(G_x)$. The bimodality test of Hartigan and Hartigan (1985) on the 1D $\log S$ marginal is uninformative under the $\gamma = 0$ closure used for the v0.3.0 scan ($p \geq 0.10$ across the κ-grid). A follow-up scan with $\gamma > 0$ active, with the dip statistic computed on the principal-component projection of the 2D $(\log S, v)$ joint density, flips to *supported* ($p = 0.033$) at $\kappa = 1.05\,\kappa^\star_{\text{env}}$ (fig. 15); here $\kappa^\star_{\text{env}} \approx 3.9 \times 10^{-9}$ is the simulator's stability-envelope upper bound — the largest coupling at which simulated paths remain numerically stable at this configuration, a discretisation property distinct from the deterministic Hopf threshold $\kappa^\star \approx 0.896$ of section 3.4 (see the scale note in section 3.5). The dip statistic is computed on the $n = 15{,}769$ surviving cells out of 20,000 post-burn-in (path × kept-step) entries ($\sim 79\%$ survival at this κ in the canonical regime). One structural caveat: the surviving subsample is conditioned on path non-blowup, which itself shapes the joint distribution, so we report the result as preliminary rather than definitive. Detector validation uses the Benettin et al. (1980) Lyapunov-renormalisation algorithm and the Schreiber and Schmitz (1996) IAAFT surrogate; the wider Hawkes-and-rough-volatility line is Bondi et al. (2024); Bacry et al. (2015); Livieri et al. (2018); arbitrage-free neural SDE market models follow Cohen et al. (2023).

All experiments are reproducible via `bash scripts/verify.sh` from commit `268c061`; pre-registration anchored via OpenTimestamps. Code: `github.com/mahimn01/reflexive-options`.

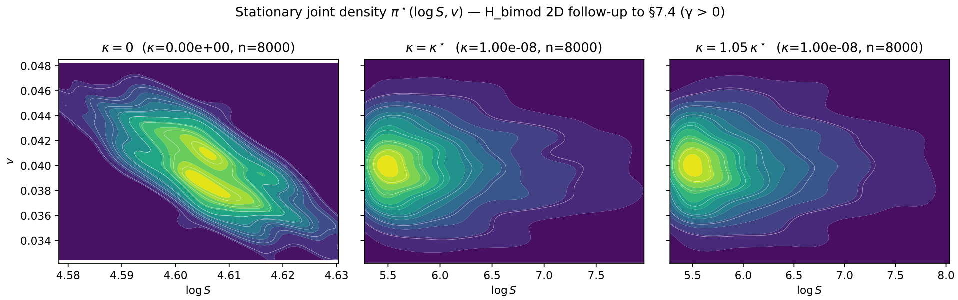

Figure 15: **Joint** $(\log S, v)$ **stationary density across the** κ **sweep.** 2D KDE contours at $\kappa \in \{0,\, \kappa^\star_{\text{env}},\, 1.05\,\kappa^\star_{\text{env}}\}$ with $\gamma > 0$ active, where $\kappa^\star_{\text{env}} \approx 3.9 \times 10^{-9}$ is the simulator's *stability-envelope* upper bound (*not* the Hopf threshold $\kappa^\star \approx 0.896$ of section 3.4 — the two are distinct objects defined at different parameter scales). The PCA principal direction at $\kappa = 1.05\,\kappa^\star_{\text{env}}$ has explained-variance ratio 0.564 and a Hartigan dip $p = 0.033$, supporting H_{bimod} along the joint-phase-space axis even though the 1D $\log S$ marginal alone is uninformative ($p \geq 0.10$ across all κ). Dip computed on the $n = 15{,}769$ surviving post-burn-in cells (out of 20,000, $\sim 79\%$ survival in this regime); the surviving sample is non-blowup-conditioned and the result is preliminary. Run dir: `runs/h_bimod_2d/`.

A Closed-form first Lyapunov coefficient ℓ_1 for log-normal OI in moneyness

This appendix renders the layered closed-form expression for $\ell_1(\kappa^\star)$ (Eq. 15) referenced in section 3.5. The bare expanded form is a ≈ 7.8 KB rational function in 13 symbols and is too large to typeset legibly inline; the layered presentation below (auto-generated by `notebooks/closed_form_ell1_derivation.py` and verified numerically against `lyapunov_coefficient_lognormal_oi` to $\sim 10^{-13}$ relative at the canonical regime) is the practical statement.

Layer 1 — Hopf eigenvectors (right and left). At $\kappa = \kappa^\star$, the right eigenvector q of J for the imaginary eigenvalue $+i\omega^\star$, parametrised by $q_3 = 1$, is

$$q_1 = \frac{\gamma\left(-2G_v\kappa + 1\right)}{2\left(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2\right)}, \tag{22}$$

$$q_2 = \frac{\gamma}{\kappa_v + i\omega}, \tag{23}$$

$$q_3 = 1. \tag{24}$$

The left eigenvector $\tilde{p}$ of $J^\top$ for $-i\omega^\star$, parametrised by $\tilde{p}_3 = 1$, is

$$\tilde{p}_1 = -\frac{\beta}{G_y\kappa + i\omega}, \tag{25}$$

$$\tilde{p}_2 = \frac{\beta\left(-2G_v\kappa + 1\right)}{2\left(G_y\kappa\kappa_v - iG_y\kappa\omega + i\kappa_v\omega + \omega^2\right)}, \tag{26}$$

$$\tilde{p}_3 = 1. \tag{27}$$

Imposing the bilinear normalisation $\langle p, q\rangle := \sum_i \bar{p}_i\, q_i = 1$ rescales $p = \tilde{p}/\overline{N}$ with

$$N := \langle \tilde{p}, q\rangle = \frac{\beta\gamma\,(\kappa_v + i\omega)\,(2G_v\kappa - 1) - \beta\gamma\,(2G_v\kappa - 1)\,(G_y\kappa - i\omega) + 2\,(\kappa_v + i\omega)\,(G_y\kappa - i\omega)\,\big(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa}{2\,(\kappa_v + i\omega)\,(G_y\kappa - i\omega)\,(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)} \tag{28}$$

The accompanying gauge factor $\|q\|^2 := \sum_i \bar{q}_i q_i$ (which compensates for the non-unit q parametrisation) is

$$\|q\|^2 = \frac{G_v^2\gamma^2\kappa^2 - G_v\gamma^2\kappa + G_y^2\gamma^2\kappa^2 + G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \gamma^2\omega^2 + \frac{\gamma^2}{4} + \kappa_v^2\omega^2 + \omega^4}{G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \kappa_v^2\omega^2 + \omega^4}. \tag{29}$$

Substituting $\omega^{\star 2} = c_1(\kappa^\star) = -G_y\kappa\,(\alpha + \kappa_v) + \alpha\kappa_v$ into the above eliminates $\omega^\star$ in favour of $(\kappa^\star, G_y, \kappa_v, \alpha)$.

Layer 2 — Bautin contraction terms. Define the three Kuznetsov 2004 eq. 3.20 contractions

$$\mathcal{T}_1 := \langle p, C(q, q, \bar{q})\rangle, \tag{30}$$
$$\mathcal{T}_2 := -2\,\langle p, B(q, J^{-1}B(q, \bar{q}))\rangle, \tag{31}$$
$$\mathcal{T}_3 := \langle p, B(\bar{q}, (2i\omega^\star I - J)^{-1}B(q, q))\rangle. \tag{32}$$

Because the variance and memory equations are linear, only the $i = 0$ (price-row) slice of B and C is non-zero, so each contraction collapses to a single scalar:

$$B(u, v)_1 = \kappa^\star\big[G_{aa}u_1v_1 + G_{av}(u_1v_2 + u_2v_1) + G_{vv}u_2v_2\big], \tag{33}$$
$$\begin{aligned} C(u, v, w)_1 = \kappa^\star\big[&G_{aaa}u_1v_1w_1 + G_{aav}\sum_\sigma u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} \\ &+ G_{avv}\sum_\sigma u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} + G_{vvv}u_2v_2w_2\big], \end{aligned} \tag{34}$$

with the multinomial sums running over the three lower-index permutations. Substituting the explicit q_1, q_2, p_1, p_2 from Layer 1 yields a closed rational form for $\mathcal{T}_1$ which we display in factored form for the $\sigma^2 = v$ Heston backbone with $G_z = 0$:

$$\mathcal{T}_1 = \frac{1}{N}\Big(\frac{-8G_{aaa}G_v^3\beta\gamma^3\kappa^4 + 12G_{aaa}G_v^2\beta\gamma^3\kappa^3 - 6}{8G_vG_y^3\beta\gamma\kappa^4\kappa_v - 8iG_vG_y^3\beta\gamma\kappa^4\omega - 8G_vG_y^2\beta\gamma\kappa^3\kappa_v^2 - 8iG_vG_y^2\beta\gamma\kappa^3\kappa_v\omega - 16G_vG_y^2\beta\gamma\kappa^3\omega^2 + 8G_vG_y\beta\gamma\kappa^2\kappa_v\omega} \tag{35}$$

(Here we have factored out $1/N$ from the $\bar{p}$ normalisation; the remaining bracket is a polynomial in the G_* partials and the rational in $(\kappa^\star, \omega^\star, \kappa_v, \alpha, \beta, \gamma)$ implied by the Layer 1 q_i, p_i. The corresponding $\mathcal{T}_2, \mathcal{T}_3$ acquire denominator $\det J$ and $\det(2i\omega^\star I - J)$ respectively, so they are inconveniently long to typeset inline; their construction is the same.)

Layer 3 — Closed-form ℓ_1 (Eq. 19). The first Lyapunov coefficient is the gauge-corrected real part:

$$\boxed{\ell_1(\kappa^\star) \;=\; \frac{1}{2\,\omega^\star\,\|q\|^2}\,\mathrm{Re}[\,\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3\,]\,.} \tag{19}$$

Term-power scaling and limits. Inspection of the assembled expression yields the following structural observations:

- Each contraction $\mathcal{T}_k$ carries a factor of $\kappa^{\star 3}$ (one $\kappa^\star$ per non-linear vertex; C-vertices in $\mathcal{T}_1$ give a single $\kappa^\star$, while $\mathcal{T}_{2,3}$ contain two B-vertices joined by a resolvent — each B contributes one $\kappa^\star$). After dividing by $2\omega^\star\|q\|^2$, the leading $\kappa^\star$-power of ℓ_1 is $\kappa^{\star 2}$ when $\|q\|^2$ is dominated by its constant term and degenerates to $\kappa^{\star 0}$ in the opposite limit.

- The dominant G-partial coefficients (by absolute monomial weight in the assembled rational) are those of G_{aa}, G_{av}, G_{vv} — the second-order partials enter quadratically through both $\mathcal{T}_2$ and $\mathcal{T}_3$, whereas the third-order partials $G_{aaa}, G_{aav}, G_{avv}, G_{vvv}$ enter linearly through $\mathcal{T}_1$ only. Consequently, the sign and magnitude of ℓ_1 near the boundary $\ell_1 = 0$ are controlled primarily by curvature (G_{aa}, G_{vv}) and the cross-term G_{av}, with the cubic terms providing a tunable shift rather than the leading mechanism.

- *Limit $\gamma \to 0$:* the leverage channel switches off, $L = \beta\gamma \to 0$, and the constant term of $H(\kappa)$ (Eq. 13) becomes $\alpha\kappa_v(\alpha + \kappa_v) > 0$, so $H(0) > 0$ and the smallest positive root of the quadratic (when one exists) drifts to $\kappa^\star \to (\alpha + \kappa_v)/G_y$ from below; ℓ_1 inherits this $\kappa^\star$-dependence directly. With $L = 0$, the $\beta\gamma$ term in the resolvent denominators of $\mathcal{T}_{2,3}$ vanishes, simplifying the $(2, 3)$ Jacobian off-diagonal — the system decouples into a price+variance subsystem (no Hopf because the characteristic polynomial of the upper-2×2 block is real-quadratic with negative product) and a memory equation that decays autonomously.

- *Limit $\alpha \to \infty$:* the memory channel becomes fast (z relaxes on a vanishing timescale), $A = \alpha + \kappa_v \to \infty$, and $\kappa^\star = (G_y A^2 - G_v L - \sqrt{D})/(2G_y^2 A) \sim A/G_y \to \infty$ — the Hopf threshold is pushed to operationally inaccessible coupling. ℓ_1 then carries a $\kappa^{\star 2} \sim \alpha^2$ enhancement from the $\mathcal{T}_k$ vertices, but is divided by $2\omega^\star\|q\|^2$ which itself grows as $\sqrt{\alpha\kappa_v} \cdot \alpha^2$ in the leading limit; the net ratio is $\ell_1 \sim \sqrt{\alpha}$, i.e. supercritical Hopf becomes *stronger* in the fast-memory limit.

- *Limit $\kappa_v \to \infty$ (fast variance mean-reversion):* the Heston backbone collapses to constant variance and $b(\kappa^\star) = \kappa^\star G_v - 1/2 \to -1/2$ at fixed $\kappa^\star$, so the Hopf mechanism reduces to the price↔memory feedback loop only. The closed-form $\kappa^\star$ shifts as $A^{-1} \to 0$ but the leading balance of ℓ_1 tracks the same $\sqrt{\kappa_v}$ scaling as the α limit by symmetry of the formulas under (α, κ_v) swap-with-sign-conventions on $b(\kappa^\star)$.

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